PRODUCT REQUIREMENTS DOCUMENT

Songloom

A cozy musical world-weaving puzzle game

You are a tiny creature in a silent, colorless world. Sing, and your song blooms into light: every note you play paints the world permanently, and every world remembers your song. This document specifies the research, design, technology and launch plan for global release on Poki.

Target platform Poki.com — web, mobile-first

Production model Solo developer + Claude AI pipeline

Timeline Six months to submission

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WEAVE THE WORLD WITH SONG

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1. Executive Summary

Songloom is a cozy, musical, world-weaving puzzle game built natively for the web and designed for one specific outcome: global release and sustained earning on Poki, the largest curated web games platform in the world. The player controls a tiny humming creature in a silent, colorless world. Singing is the single verb of the game: every note the player chooses blooms outward as a wave of colored light that both reveals the hidden geometry of the world and permanently paints it. Different pitches do different things to the world - low notes grow roots and bridges, mid notes wake and charm creatures, high notes shatter crystal, chords open ancient doors, and silence lets the player sneak past listening creatures. The result is a game where the player composes their way through the world, and the world becomes a painting of their melody.

This document is the complete product requirements specification for Songloom. It is written for a production model of exactly one developer working with Claude as an AI production partner, and it assumes a hard budget constraint of near-zero cash cost. Every section is written to be executable: mechanics are specified to the level of control schemes and event schemas, art and audio are specified to the level of palette values and synthesis graphs, and the launch plan is specified to the level of submission checklists and metric thresholds. The document is deliberately long because its job is to remove ambiguity from a six-month solo production.

100M	$0	6 mo
monthly players on Poki (2026)	cash budget; Claude subscription only	from first commit to Poki submission

The strategic thesis is simple. Poki publicly rejects games that clone existing hits, and the platform is currently flooded with AI-assisted lookalike submissions that recycle the same mechanics, assets, and sounds. Meanwhile, the closest genre to Songloom - echolocation games where sound reveals a dark world - exists only on mobile app stores and Steam, almost always in a horror register, and never on the open web. Songloom occupies the intersection of three verified gaps: a tonal gap (cozy and colorful instead of dark and scary), a mechanical gap (pitch is the control scheme and painting is the outcome, not just a reveal), and a platform gap (web-native, portrait-friendly, tiny file size, instant play). The developer's professional strengths - sound engineering, motion graphics, VFX, 3D, and machine learning - map directly onto the game's core system, which makes the game hard to clone quickly and cheap to build well.

The revenue expectation is calibrated to public statements from Poki's own developer relations team: a realistic first web game earns between five hundred and three thousand US dollars per month from advertising revenue share. The plan targets the upper end of that band through Poki's own success doctrine - longer average playtimes, higher replay rates, and post-launch content updates that keep the game eligible for platform promotion. Virality is engineered rather than hoped for: a daily seeded puzzle shared worldwide, spoiler-free shareable result cards in the Wordle pattern, and streamer-friendly spectacle moments every twenty to forty seconds. The full financial model, risk register, and the complete Poki

submission playbook are contained in the final sections of this document.

2. Platform Research: Poki, Inside and Out

This chapter records what was verified about Poki from primary sources: the official developer documentation at developers.poki.com, the platform's own published statements, and interviews with Poki staff. Everything here is either a direct platform rule or a directly published number. Where a figure varies between sources, the more recent official statement is used. This matters because a rejection on Poki is not a bug to fix later - it is a design failure to prevent now.

2.1 Scale and business model

Poki is an Amsterdam-based company operating poki.com, a curated platform for HTML5 games playable directly in the browser with no download and no account. As of 2026 the platform reports roughly one hundred million monthly active players, more than one billion gameplays per month, and a cumulative total above six hundred and twenty-five million players since launch. The community of active developers is around five hundred, and Poki states that a high-performing game can expect more than five million monthly players on its own. The platform monetizes through advertising and shares that revenue with developers; there is no upfront fee for developers, and every testing tool described below is free.

Crucially, Poki is a curated marketplace, not an open upload site. Every game passes human review before release, and the team explicitly evaluates whether a game is unique enough within its categories. The official guidance is unambiguous on clones: games that copy existing hits do not stand out to players and will not be promoted. This single fact shaped the entire concept selection process in this document - originality is not a nice-to-have, it is the entry fee.

2.2 What Poki officially looks for

The developer documentation lists a baseline of three signals and a set of five differentiators beyond them. The baseline three are quality (assessed through UX feel and the strength of the core game loop), player fit (whether Poki's actual global audience would enjoy the game), and technical soundness (fast load times, stable frame rates, and lean file size across devices and engines). The five differentiators are the real acceptance criteria:

- Originality - a fresh idea, mechanic, or twist. Clones are explicitly named as unpromotable.
- Depth and replayability - a satisfying core loop that opens up over time, rather than a single clever idea.
- First impression - a clear thumbnail and an immediately playable opening, because players decide in seconds.
- Cross-device feel - controls and UI that work as well with a thumb as with a mouse; the audience is majority mobile.

- Room to grow - content hooks, systems that support updates, and a roadmap shared at the time of submission.

Poki adds an important nuance: none of these is a pass-fail checklist. They are weighed together, and a game that is exceptional on one dimension can compensate for being weaker on another. In practice this means a solo developer should aim to be exceptional on originality and first impression - the two dimensions most under the developer's direct control - while meeting solid baselines on the rest.

2.3 The release pipeline and its metric bars

Every game on Poki goes through the same five-step journey, and every step returns real player data that the developer may act on before moving forward. The pipeline is free and repeatable, which makes it a genuine design tool rather than just a gate.

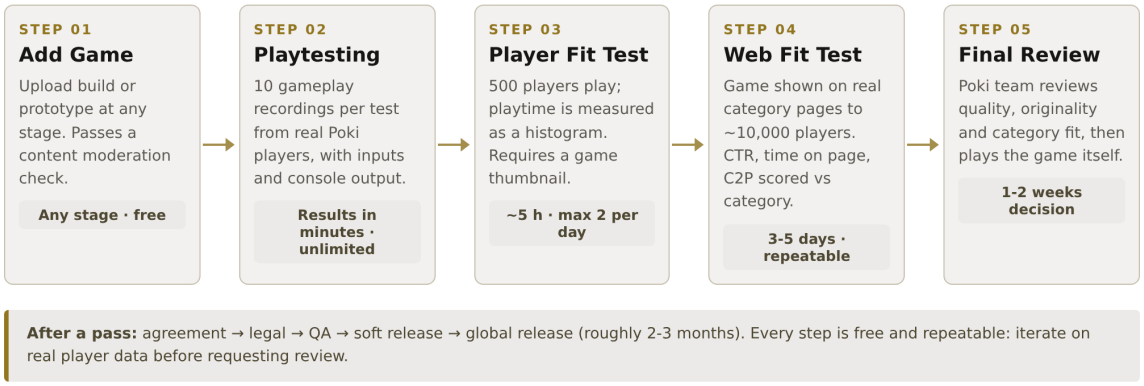


Figure 2.1 - The five-step Poki release pipeline, with durations and what each step measures.

The player fit test is the heart of the system: five hundred real Poki players play the game, and the platform returns a histogram of how long each of them played. The stated healthy result is an average playtime above three minutes with at least twenty-five percent of plays lasting longer than three minutes. The web fit test then places the game on real category pages for roughly ten thousand players and scores three metrics against category benchmarks: click-through rate, average time on page, and conversion to play. Conversion to play depends on the first gameplayStart SDK event firing, so a missing SDK integration makes the test unreadable. Poki's own guidance on what to aim for before requesting final review:

Metric	Minimum to advance	Recommended before review	Average released game
Average playtime	3+ minutes	5+ minutes	~6 minutes
Plays longer than 3 min	25% of plays	30% of plays	~35%
Conversion to play (C2P)	measured at web fit	65%+	~70%
Management / sim games	n/a	10+ minutes playtime	higher band

Table 2.1 - Poki pipeline metric thresholds (from official documentation)

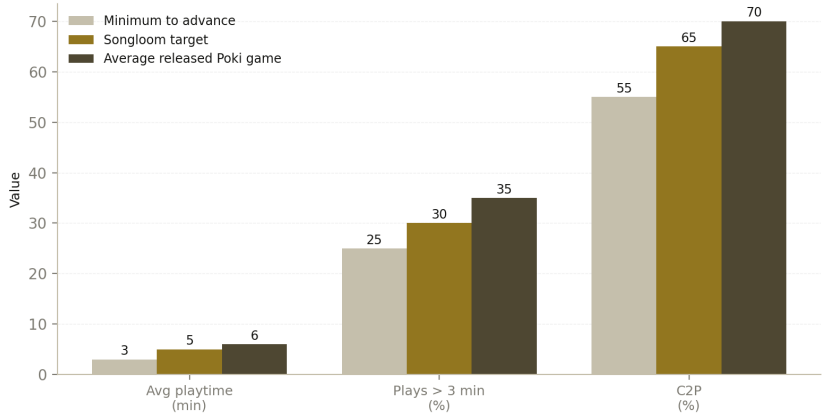


Figure 2.2 - Songloom targets against the platform minimum and the average released game.

2.4 Technical requirements and guidance

Poki's engine guide gives the clearest technical budget in web gaming: for a good web game the initial download should not exceed five megabytes and the total should stay under eight megabytes. Portrait orientation is strongly advised, for two reasons: portrait-playable games see more players actually enter gameplay on average, and they become eligible for the Gamebar Display ad format - a mobile ad placed in the title bar that earns extra revenue with no additional work from the developer. The platform reports that roughly sixty-eight percent of players have WebGPU-capable browsers as of mid-2026, and its engineers recommend treating WebGPU as an enhancement layer over a WebGL2 baseline rather than a requirement.

- Initial download: 5 MB or less. Total build: 8 MB or less. These are the platform's own numbers.
- Portrait support is a distribution advantage, not a courtesy: more gameplay entries plus Gamebar ad revenue.
- Progressive loading - menu, tutorial, and first level first, remaining content fetched in the background - is described by Poki as one of the most important techniques for web success.
- Drop-off during loading is brutally underestimated, per the official onboarding guide; a visual loading screen with a progress bar is required practice.

- Multiplayer games face thin test traffic; Poki itself recommends bots to fill lobbies during tests. Songloom is single-player by design, sidestepping this entirely.

2.5 The engagement doctrine

Poki's engagement documentation is unusually specific, and it reads like a design brief for exactly the kind of game Songloom is. Web players have shorter sessions and return less often than mobile players, so the first session carries most of the weight. The platform recommends quick, satisfying, repeatable loops of around three minutes, and roughly one hour of total content as the right scope - sprawling progression systems rarely translate to web, and memory limits how much fits in a small build anyway. The stated philosophy: make the first minutes brilliant before making the tenth hour deep.

The onboarding doctrine is even more direct. Skip menus for first-time players and put them straight into the good part. During the first three minutes the player should not be able to die or fail - Poki's own developer relations staff say hand-holding at the start is not a design weakness on web, it is a necessity. Explain with visuals rather than text walls, because English text excludes a large share of the global audience. Reveal controls gradually across the first levels instead of front-loading everything. Games where the player can constantly perform an action outperform games with waiting and downtime, across every genre. And congratulate the player at every milestone - confetti, a happy sound, a few words of praise.

2.6 Monetization doctrine

Monetization on Poki is advertising revenue share, and the platform's own tips document prescribes how to design for it. The economy should be simple: one understandable currency, clear rewards, and no secondary currencies, which web players dislike. Rewarded video placements work in three families - a helping hand for struggling players (revives, level skips, hints, boosts), in-game economy choices (spend coins or watch a video for the same reward), and customization ownership. Best practices: the non-rewarded option always sits in the primary position, never oversell the rewarded button, use the industry-standard video icon, celebrate the reward with animation and sound, and prefer dynamic contextual placements over static buttons. Weekly and seasonal unlocks lift engagement and retention. All of these are adopted verbatim in Chapter 17.

2.7 Timeline and post-release

Final review takes one to two weeks. A pass leads to agreement, legal, platform QA, soft release, and then global release, roughly two to three months in total. After release the doctrine is equally clear: games that keep improving keep earning promotion. Bug fixes and performance lifts make a game eligible for more promotion; content updates attract new players and bring existing ones back; adding languages directly grows the reachable audience for text-heavy games. Songloom is designed text-light and update-heavy by default, which is covered in Chapter 20.

3. Market Analysis: The Web Game Opportunity

The browser games market is small relative to mobile and console, but it is growing steadily and its distribution economics are uniquely favorable to a zero-budget solo developer. According to The Business Research Company, the global browser games market sits at approximately eight billion US dollars in 2026 and is on track to reach roughly nine point one billion by 2030. That is a real commercial market, not a hobby scene - and unlike app stores, it requires no developer fees, no review queues measured in days, and no user acquisition spend to be discoverable, because the platform itself is the store.

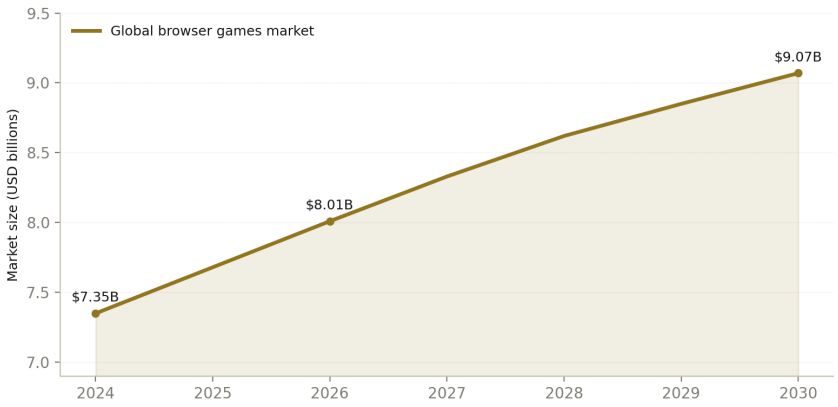


Figure 3.1 - Global browser games market size, 2024-2030 (The Business Research Company).

3.1 Who the web player is

Poki's 2026 State of Web Gaming study surveyed two thousand gamers and four hundred developers, and its findings define the audience Songloom is built for. Thirty-seven percent of surveyed consumers play web games, and they are not low-value players: twenty-seven percent of web gamers spend more than fifty dollars per month across the gaming ecosystem, rising to thirty-five percent among the most frequent players. Fifty-three percent of surveyed developers see web games as a way to reach new users, and a majority of players actively choose the browser channel. The platform's own commentary adds the behavioral layer: the audience is younger, raised on short-form content and instant access, and many players arrive through a link or a search still deciding whether to commit. The game has to earn that commitment in seconds.

- Instant gratification is baseline: click, then play within seconds, or the player leaves before the title screen.

- Session shape is 2-5 minutes of easy-to-start value, repeated many times, rather than one long committed sit.
- The creativity segment is large and underserved: Poki states that its girl-skewing audience responds strongly to self-expression, customization, and open creative spaces with no win condition.
- Global by default: Poki serves more than thirty languages; visual, text-light design travels without localization cost.

3.2 The cozy wave

Across the wider industry, the fastest-growing player motivation is the one Songloom is built around. Cozy gaming moved from a niche label to a mainstream category as players increasingly choose games that help them unwind, improve mood, and offer low-pressure play: atmosphere, intimacy, and creativity instead of combat and pressure. Industry commentary describes gentle non-violent world-building as one of the clearest trends of the decade, growing across regions and strongest among younger players who use games to manage stress. On Poki specifically, the platform's own audience insights - creativity, self-expression, accessible difficulty, no-win-condition play spaces - read as a description of a cozy game audience that the platform's current catalog serves mostly through dress-up and coloring formats. There is very little cozy content with a genuine game underneath it. That is the gap.

3.3 The AI-slop problem, and why it is an opportunity

The industry's 2026 conversation is dominated by a flood of low-effort, AI-generated games - the press regularly covers AI slop filling charts and platforms with lookalike product. Submissions to web platforms increasingly recycle the same templates, the same assets, the same color schemes, and the same loop structure. Players and platform curators have noticed, and curation has tightened in response. For a developer willing to do real craft, this is not a threat; it is a moat. When the median submission is generic, a game with a genuinely original mechanic, hand-tuned art direction, and a distinctive audio identity does not merely compete - it stands out by default. Poki's own originality criterion effectively prices the flood out: clones are explicitly unpromotable.

Songloom's positioning against the flood is deliberate. Its audio is synthesized from scratch rather than assembled from stock packs - the developer's sound engineering background makes this the cheapest asset class to produce at high quality, while being the hardest thing for a prompt-and-template workflow to fake. Its visual identity is a specific, opinionated two-layer system rather than a generic palette. Its interaction model does not exist in any template. In a market saturated with sameness, specificity is the strategy.

3.4 Platform choice: why Poki first

The web has several distribution platforms - Poki, CrazyGames, Y8, itch.io, and self-publishing. Poki is the target for the first release for concrete reasons rather than habit: it has the largest audience in the segment, the most developed testing infrastructure (real player playtests and fit tests before any release decision), an

explicit long-term partnership posture (promotion, QA, brand protection), and the clearest published quality doctrine, which reduces the risk of building the wrong thing. A secondary benefit is reputational: a game that passes Poki's curation carries a quality signal that eases later distribution elsewhere.

Platform	Model	Fit for Songloom	Notes
Poki	Curated, ad revenue share	Primary target	100M MAU; full testing pipeline; long-term promotion posture
CrazyGames	Curated, ad revenue share	Secondary after launch	Comparable model, large audience; submission after Poki soft release
Y8 / GameDistribution	Semi-open distribution	Optional later	Broad reach, less curation, lower promotion support
itch.io	Open upload	Devlog + beta builds	Community feedback channel, not a revenue channel
Self-hosted site	Direct	Marketing asset only	Press kit, demo, and devlog home; no organic traffic

Table 3.1 - Web game platform comparison for a first release

4. Competitive Landscape and the Whitespace

This chapter maps what already exists, what is oversupplied, and exactly where the open space sits. The analysis covers the current Poki catalog, the adjacent echolocation genre, and the viral web game tradition that Songloom draws its distribution mechanics from. The purpose is defensive: the single most important commercial risk named in the brief is rejection for being insufficiently unique, and this chapter is the evidence that the concept is not a clone of anything already published.

4.1 What is oversupplied on Poki

Direct catalog inspection shows entire genres at saturation. The platform's most popular pages are dominated by io arena games, stickman platformers, endless runners, idle and merge games, tower defense, shop and restaurant management sims, car and drift games, dress-up and coloring formats, and horror escape rooms. Each of these categories already has multiple entrenched, heavily promoted incumbents - Subway Surfers and Temple Run in runners, Monkey Mart in management, Level Devil in troll platformers, Stickman Hook in skill swinging, the entire io shelf. A new entrant in any of these categories is not competing on quality; it is competing against incumbents with millions of accumulated players and platform promotion history. The saturation map below guided concept selection: anything scoring eight or above was excluded from consideration.

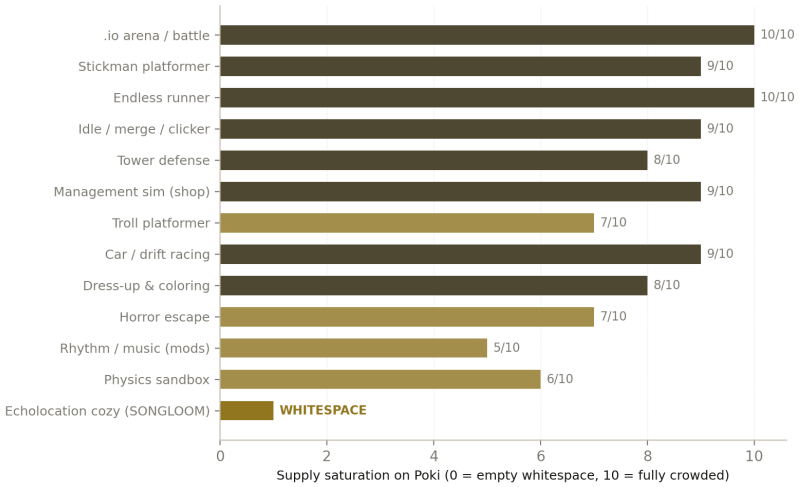


Figure 4.1 - Supply saturation by genre on Poki, from direct catalog inspection (10 = fully crowded).

4.2 The closest neighbors: echolocation games

The closest existing genre to Songloom is the echolocation game, where sound made by the player reveals an otherwise invisible world. The genre has real precedents, and this document treats them seriously rather than ignoring them. Dark Echo (2015, mobile) is the progenitor: footsteps and thrown objects emit visible sound lines that outline the level, and monsters hear the player. Auris (Steam) iterates on the same idea with sound waves revealing platforms and hazards. Echo Maze and its variants (2026, mobile app stores) have players clap to send echo waves through dark mazes. Perception, Blind, Sounds of Shadows, and EchoBlade explore the same space on PC and console, typically with blind protagonists and horror atmosphere.

Game	Platform	Tone	Mechanic	Gap left open
Dark Echo	iOS / Android	Horror / tense	Footsteps emit outline lines; monsters hear you	Web, color, painting, cozy tone
Auris	Steam	Dark / serious	Sound waves reveal platforms and hazards	Web, pitch-verbs, permanence
Echo Maze family	iOS / Android	Horror	Clap to reveal maze briefly	Web, depth, replay systems
Perception / Blind	PC / console	Horror narrati ve	Sonar pings reveal environment	Web, sessions, file size
Sounds of Shadows	PC	Atmospheric	Sound waves guide a blind robot	Web, creativity segment

Table 4.1 - Closest-genre competitor teardown

The pattern is unambiguous: every game in the genre is dark, tense or horror-toned, distributed through app stores or Steam, and uses sound as a torch - a binary reveal of pre-existing geometry. None is on the open

web. None is cozy. None uses pitch as a control scheme with different world verbs. None treats the revealed world as a persistent painting the player creates rather than a temporary flashlight sweep. The genre has been thoroughly explored in one narrow register, and completely ignored in every other register.

4.3 The whitespace thesis

Songloom's uniqueness claim is therefore precise rather than vague, and it is built from three independent gaps that compound when combined. First, tonal: the entire echolocation space is dark; the cozy space on Poki is dominated by shallow formats; nothing combines genuine puzzle depth with a warm, low-pressure register. Second, mechanical: in existing games sound reveals the world; in Songloom sound paints and rewrites the world, and which pitch you choose is the strategic decision - the verb system documented in Chapter 6 does not exist in any shipped game found in this research. Third, platform: the web-native constraints Poki rewards - instant play, tiny builds, portrait, one-thumb controls, three-minute loops - are exactly the constraints the incumbents' engines and store distribution ignore.

> *The claim is not that no game has ever made sound visible. It is that no game has made sound the brush, the world the canvas, the pitch the player's choice, and the painting permanent - and no such game runs in a browser tab in three megabytes.*

4.4 Lessons from the viral tradition

Songloom borrows distribution mechanics, not gameplay, from the canonical viral web games. From Wordle: one puzzle per day, identical worldwide, with a spoiler-free share format that shows the journey rather than the solution - the pattern analysts credit as the single most important growth mechanic of the biggest web game phenomenon of the decade. From the io wave: simplicity of spectacle, where the game is visually legible in a two-second clip. From Friday Night Funkin: a rhythm game that became a global phenomenon from a browser demo because it was built on a distinctive audio-visual identity that the community could remix. From Suika: physics-toy satisfaction with a collection hook. From Level Devil, the most recent Poki-native breakout: humor, single-mechanic clarity, and a solo developer with a voice. The common thread is that every one of these games was instantly explainable in one sentence and one image. Songloom passes the same test: you sing, and the world blooms.

5. The Concept: SONGLOOM

Songloom is a cozy musical world-weaving puzzle game for the web. The player is a Hummy - a small, round, lantern-hearted creature - awakened alone in the Loom, a world that has fallen silent and drained of color. The Hummy carries the last song. Every note it sings rolls outward as a bloom wave that reveals the world's hidden geometry and paints it in the note's color, permanently. As the player sings their way through the Loom's chambers, the world regrows, its creatures wake, and the player's own melody becomes the record of their journey: the world literally remembers your song.

1 verb singing - the entire control scheme	**5 voices** low, mid, high, chord, silence	**0** audio files in the build - all sound is synthesized

5.1 The pitch-to-verb system

The core originality is that pitch is the control scheme. When the player holds to sing, dragging vertically selects a pitch, and each pitch band is a different world verb with its own color. Low notes grow. Mid notes charm. High notes shatter. Chords open. Silence hides. Because each verb is paired with a color, the painted world becomes a readable record of the player's decisions, and every puzzle is an invitation to think musically: which voice does this place need? The verb table is small enough to fit in one glance and deep enough to generate hundreds of chambers, which is exactly the shape a web hit needs - seconds to grasp, weeks to master.

Voice	Pitch · Color	World Verbs	Signature Use
Root Hum	Low · Moss	**Grow** vines, roots and bridges · **Feed** Bloomlings · **Soften** soil mounds	Raise a root bridge across the Hollow gap; moss spreads wherever the wave passes.
Lantern Tone	Mid · Amber	**Wake** and charm creatures · **Light** lanterns · **Attract** Hushlings (stealth risk)	A sleeping Bloomling hums back and carries your note to a far switch.
Star Chime	High · Sky	**Shatter** crystal · **Reveal** hidden glyphs · **Charge** echo crystals	A single chime cracks the crystal dam; the water of the chorus flows again.
Chord	Multi · Iris	**Open** Bloom Doors · **Harmonize** duet stones · **Resolve** puzzles needing two colors at once	Hold two voices at once (two-thumb drag) to wake the Elderchoir door.
Silence	None · Faded	**Sneak** past Hushlings · **Let** blooms fade to reset a pattern · **Listen** to the world's own song	Release the note and go quiet: the Hushling drifts back to sleep, the path clears.

> The verb table is the whole control scheme: **pitch is the input, color is the feedback, verbs are the outcome**. Everything the player can do is a combination of these five voices. Swatch colors are the in-game bloom palette targets, shown here as neutral approximations for print.

Figure 5.1 - The pitch-to-color-to-verb table: the entire control scheme of Songloom on one page.

5.2 The fantasy and the emotional arc

The fantasy is restoring a world with your own voice. The arc of a session moves from silence to music: the player enters a colorless chamber, reads its shapes, chooses their first note, and watches the room come alive in a cascade of color and sound. Failure does not exist as punishment - a wrong note simply fades, and the world waits. Mastery is expressed as beauty and efficiency: fewer notes, fuller painting, higher resonance. The end state of a finished chamber is a quiet, fully painted garden that humns with the player's own melody. This is the register the cozy audience is spending its time seeking, and it is the emotional inverse of every horror-toned echolocation game on the market.

5.3 Why the developer is the moat

The concept was selected in part because it is built on the developer's rarest skills rather than in spite of the budget. The game's audio is entirely procedural synthesis through the Web Audio API - no sample packs, no licensing, no asset weight - and audio quality is the game's identity. A professional sound engineering background makes that both free and premium, while it remains the hardest thing in the game for a template-driven AI workflow to imitate credibly. The visual identity - ink-line geometry that saturates under bloom waves - leans on motion graphics and VFX craft, the second-strongest skill in the stack. Machine learning experience feeds the production pipeline: analytics interpretation, playtest recording triage, and difficulty tuning tooling described in Chapters 23 and 24. In short: the game is engineered so that the expensive-looking parts are the cheap parts for this specific developer.

5.4 Design pillars

- **Sound is vision.** Everything the player knows about the world, they learn by singing. If a mechanic cannot be heard or seen blooming, it does not ship.
- **No wrong notes early.** The first minutes are consequence-free by doctrine; the game earns trust before it asks for skill (Poki onboarding rule, adopted verbatim).
- **The world remembers.** Painted regions persist. The player's session leaves a visible, screenshot-able artifact - the bridge between gameplay and sharing.
- **One thumb, five voices.** Fully playable - and fully enjoyable - held one-handed on a bus. Depth comes from pitch choice and timing, never from control complexity.
- **Small file, big wonder.** Three megabytes initial download, everything synthesized or generated. Load in seconds, bloom in one.

5.5 Positioning statement

> *For players who love cozy worlds and clever puzzles, Songloom is the game where your voice paints the world back to life - unlike any dark echo game on the app stores or any coloring format on the web, because here the music is the brush, the puzzle is the score, and the world remembers your song.*

6. Core Gameplay and Mechanics

This chapter specifies the game to the level of inputs, timings, and numbers. Every mechanic below is designed against the platform doctrine established in Chapter 2: three-minute loops, no early failure, constant action, one-thumb play, and a session zero that must be brilliant on its own.

6.1 Movement

The Hummy moves by tap-to-hop. A tap anywhere on the walkable ground queues a hop to the nearest valid anchor point (ledges, roots, stones, platforms are all anchor-tagged in the chamber data); a held drag draws an arc preview for longer hops between distant anchors, with the arc's reach capped so routing stays a decision. Hops are weighty and squashy - 0.4 seconds including anticipation and landing recovery - so movement itself feels musical. The player can hold while hopping: singing and moving overlap freely, which keeps the action continuous, per the platform's constant-action rule. On desktop, WASD or arrow keys hop between anchors and the mouse replicates every touch gesture. There is never a moment where the correct action is to wait passively; even while a puzzle resolves, the player can hum quietly to keep painting.

6.2 Singing - the primary verb

Press and hold anywhere to sing. While held, a soft note sustains and vertical drag sweeps pitch through the Hummy's range - roughly two octaves, mapped to the five voices. The note's bloom color previews live on the Hummy's lantern heart. Release to let the bloom wave roll out from the Hummy: a ring of colored light that expands for up to 4 seconds, revealing and painting everything it passes. Quiet taps produce short chirps (fast, small radius, useful for stealth-lite); charged holds produce long waves (slow, huge radius, full paint). The wave's audio is the note itself with a delay tail, so what the player hears and what the world sees are the same event - the game's single most important feel property.

Input	Result	Wave radius	Use
Tap (touch / mouse / space)	Chirp: short note, small wave	2.5 tiles	Quick reveals, gentle pings, low noise
Hold + release (no drag)	Sustained note at current pitch	7 tiles	Standard painting and puzzle solving
Hold + vertical drag	Pitch sweep while singing	7 tiles	Choosing the verb; performance expression
Two-finger hold (or shift-drag)	Chord of two pitches	6 tiles, dual color	Bloom Doors, duets, expert puzzles
Release all, stay still 1.5 s	Silence state	n/a	Sneak past Hushlings, let paint fade

Table 6.1 - Singing input specification

6.3 The bloom wave - reveal and paint

The bloom wave is one system doing two jobs. Reveal: geometry touched by the wave gains its ink outline and interior shading, so the player can read walls, gaps, and objects - the classic echolocation payoff. Paint: at the wave's trailing edge, touched surfaces take on the note's color at a saturation level proportional to how long the wave overlapped them, and the paint persists for the rest of the chamber. Objects with verb responses react on contact with the wave, not the paint: roots grow when a Root Hum wave reaches them, a Bloomling wakes when Lantern Tone touches it. Painted ground additionally functions as resonant territory - see 6.4. Wave physics is deliberately simple: constant speed, no reflection, slight vertical squash, because the puzzle language must stay legible rather than physical.

6.4 Resonance

Each chamber tracks a Resonance meter, which is the player's score expression and the game's replacement for health. Resonance rises when waves pass over unpainted ground (new paint), when creatures are charmed, and when puzzle steps resolve; it slowly decays during silence. High resonance at chamber end pays bonus Harmony Notes and unlocks the melodic finale layer of the adaptive score. Resonance never gates progress - it only rewards fuller, more musical play, which keeps the cozy contract intact while giving

skilled players a real optimization surface for stars, streaks, and leader-adjacent Daily Loom results.

6.5 Creatures

Creature	Role	Behavior	Verb interaction
Hummy	Player avatar	Hops, sings, paints	n/a
Bloomling	Helper / tool	Sleeps until woken by Lantern Tone; follows music; carries your note to remote switches	Mid notes wake and charm
Hushling	Gentle obstacle	Drifts toward sustained sound and blocks paths by absorbing waves; sleeps in silence	Silence is safety; chirps are too short to wake it
Echo Moss	Repeater / relay	Re-emits any wave that touches it, in the same color, from its own position	All voices; routing tool
Elderchoir	Gatekeeper	Dormant door-statue that opens only for chords and duet sequences	Chords; late-game duets
Thornvine	Soft hazard	Stirs when Root Hum is sung nearby, briefly blocking a lane - a routing puzzle, not a punishment	Low notes; timing

Table 6.2 - Creature roster and behaviors

The Hushling deserves a note because it carries the game's entire stealth vocabulary, and it is deliberately the softest possible version of an enemy. It never harms the Hummy; it merely drifts toward singing and settles where it absorbs waves, so a careless singer slowly loses workspace. The counterplay - go quiet, watch where it settles, route around it - produces exactly the tension-then-relief rhythm that makes echolocation games compelling, with none of the punishment that would break the cozy contract and the no-early-failure doctrine.

6.6 The no-fail doctrine

There is no death, no game over, and no fail state anywhere in Songloom. A chamber that is going badly can be re-entered freely; paint resets, progress is kept, nothing is lost except the opportunity to have painted it more fully. Consequence exists as texture, not penalty: Thornvines close a lane for seconds, Hushlings drift into inconvenient spots, crystal that shattered early may have been needed later - and the chamber simply asks the player to think again. This is not a simplification for its own sake; it is the platform's stated onboarding requirement elevated to a design philosophy, and it is also what makes the game watchable and relaxed on streams, where fail-states read as stress rather than content.

6.7 Session anatomy - the three-minute loop

A single chamber is a complete three-minute experience: enter in silence, read the space, sing the first exploratory wave, understand the puzzle, execute two to five verb decisions, resolve the goal, and receive the

celebration bloom plus resonance payout. Chambers chain through the biome overworld map with one tap. The first session target - the one that determines player fit test results - is four chambers: tutorial, two early verb lessons, and one moment of genuine surprise, ending in the first Bloom Garden glimpse. Every mechanic above is specified so that this first session contains zero waiting, zero text walls beyond single-word labels, and zero moments where the correct action is unclear.

6.8 The first five minutes, minute by minute

Because session zero carries most of the weight on the web, it is scripted to the second. The loading card appears in under one second from navigation (stage-1 bundle), and its first tap is both the audio unlock and the gameplayStart event. The Hummy wakes mid-chamber with no menu and no logo beyond the ink wordmark on the loading card. The first minute is pure promise: one tap produces the first chirp and the first sliver of revealed world; the game holds still for two full seconds and lets the wave be beautiful. The second minute teaches the held note: the lantern heart pulses amber near mid pitch, the sleeping Bloomling stirs at the wave's edge, and waking it paints the room's first color and opens the door. The third minute introduces gentle consequence: a second chamber with one root stump and a visible gap, teaching growth through the same self-confirming loop. The fourth minute delivers the surprise chamber - moss that relays a note around a corner, the first aha - and the fifth ends on the celebration cascade, the Harmony Notes count-up, a happyTime moment, and the postcard-style overworld appearing for the first time with the Daily Loom banner glowing. Five minutes, four chambers, two verbs, one aha, zero failure, one shareable moment already on screen.

7. Content Architecture

Songloom's content is structured as five biomes of eight song chambers each - forty chambers of hand-authored puzzle design - plus the Daily Loom and the Bloom Garden. The structure is sized exactly to the platform guidance: roughly one hour of core content in satisfying three-minute units, with systems that extend replay far beyond the authored hour.

7.1 The five biomes

#	Biome	New verb	New creature	Puzzle identity	Chambers
1	Whisper Hollow	Mid - Lantern Tone (wake, charm)	Bloomling	Reveal and light: reading dark rooms, waking helpers	8 (1-4 are no-fail by design)
2	Rootside Glen	Low - Root Hum (grow, bridge)	Thornvine	Growth routing: growing paths at the right moments	8
3	Chimefall Terraces	High - Star Chime (shatter, charge)	Echo Moss	Destructive precision: breaking the right crystal, relays	8
4	The Hushed Maze	Silence (sneak, fade, listen)	Hushling	Stealth-lite and pattern resets under listening pressure	8
5	Elderchoir Keep	Chord (open, harmonize, duet)	Elderchoir	Two-voice mastery: chords, duet sequences, finales	8

Table 7.1 - Biome progression: each biome introduces one verb and one creature

The biome order is the difficulty ladder and the teaching order at once. The player meets the friendliest verb first - waking helpers with a warm mid tone - because it produces the strongest immediate emotional payoff: something in the world responds to you. Growth comes second because it visibly rewrites the room. Shattering comes third, once the player trusts their voice and will enjoy precision. Silence arrives fourth as a change of texture rather than a difficulty spike. Chords finish the arc by combining everything. Biome one chambers one through four are mechanically incapable of being failed or soft-locked, per the no-early-failure doctrine; even in biome five, setbacks cost seconds, never progress.

7.2 The Daily Loom

The Daily Loom is one extra chamber per day, generated from a deterministic seed so that every player on Earth receives the identical puzzle each day. It uses a constrained generator - a validated template grammar with seeded placement, not raw randomness - so every daily chamber is guaranteed solvable, fair, and interesting (generation and validation are specified in Chapter 12). The daily carries a streak counter, a three-star grading (notes used, time, paint coverage), and produces the Song Card share image. It is the retention engine and the virality engine at once: the Wordle pattern, adapted to a world where the share artifact is a painting of your own melody rather than a grid of letters.

7.3 The Bloom Garden

The Bloom Garden is an open creative space with no goal, no timer, and no fail possibility: a single evolving garden canvas that the player paints and replants over weeks. Unlocked after the first biome, it persists between sessions and becomes the player's own corner of the Loom. It exists because the platform's own audience research says the creativity and self-expression segment - including its large girl-skewing audience - responds most strongly to exactly this: open creative space with no win condition. It also serves commerce

honestly: garden ornaments and instrument cosmetics are the game's cosmetic economy, placed where they are wanted rather than pushed.

7.4 Collection and long-term goals

- Melody Shards - 96 hidden fragments across chambers; each completes a song in the player's Songbook (30+ short melodies, playable in the Garden).
- Instrument voices - 12 unlockable cosmetic timbres for the Hummy's voice (Bell, Reed, Glass, Wood, Choir, and so on); purely timbral, never mechanical.
- Paint palettes - 8 alternate bloom palettes earned through biomes, dailies, and seasonal events.
- Stars - three per chamber (notes, time, coverage); the skilled-player surface that feeds speedrunning communities.
- Streaks - Daily Loom attendance with repair mechanics that deliberately use the rewarded ad slot (Chapter 17).

7.5 The forty chambers

The complete launch inventory, chamber by chamber. Each row names the chamber, its dominant pattern, and its single twist - the aha beat that makes it worth playing. This table is the production checklist for months three through five: a chamber is done when its twist works in playtest, its stars are reachable, and its celebration lands.

Biome	Chamber	Name	Pattern	The twist (aha)
1 Hollow	C1	The First Song	Reveal	The world answers your very first chirp
1 Hollow	C2	Sleepy Bell	Charm	The Bloomling hums your note back, an octave up
1 Hollow	C3	Paper Walls	Reveal	One wall is paint, not ink - it fades when sung
1 Hollow	C4	Two Lanterns	Charm	Both must be woken in one wave; the second is behind the first
1 Hollow	C5	Hollow Choir	Sequence	Three sleeping Bloomlings, woken in low-to-high order
1 Hollow	C6	The Listener	Stealth-lite	First Hushling: it settles where you sang, so sing elsewhere
1 Hollow	C7	Fern Lattice	Paint-match	Door seals read two colors; the room teaches both

Biome	Chamber	Name	Pattern	The twist (aha)
1 Hollow	C8	Hollowfall	Boss-choir lite	The biome door answers a held chord
2 Rootside	C9	First Root	Grow	One stump, one gap - the lesson, gently
2 Rootside	C10	Thin and Thick	Grow	The spindly bridge sags; strength was a choice
2 Rootside	C11	Two Stumps	Grow + Relay	Grow the far stump, then moss carries the charm
2 Rootside	C12	Thorn Lane	Timing	Thornvines stir near low notes - route around, then grow
2 Rootside	C13	The Stair Garden	Restore	Faded paint must be re-sung before moss wakes
2 Rootside	C14	Glen Deep	Grow + Charm	A Bloomling farms for you: sing, it plants, you grow
2 Rootside	C15	Undergrove	Reveal + Grow	The room is upside down; roots grow up
2 Rootside	C16	Greenfall	Boss-choir	Roots and charm resolve together in one cascade
3 Terraces	C17	First Chime	Shatter	One crystal, one dam - the payoff of precision
3 Terraces	C18	Four Dams	Shatter	Only the cracked one opens the waterway
3 Terraces	C19	Moss Steps	Relay	Chain three moss pads around a spiral wall
3 Terraces	C20	Chime Echo	Relay + Shatter	Shatter only what the moss can reach
3 Terraces	C21	Glasshouse	Paint-match	Terrace seals want Sky over Amber - inverted teaching
3 Terraces	C22	The Long Note	Sequence	One held wave, three relays, one perfect line
3 Terraces	C23	Overgrown Chimes	Restore + Shatter	Shattering overgrowth reveals the chimes beneath
3 Terraces	C24	Skyfall	Boss-choir	Water, light, and glass in the biome finale
4 Maze	C25	Quiet Paths	Stealth	Two Hushlings, one song: place your noise

Biome	Chamber	Name	Pattern	The twist (aha)
4 Maze	C26	The Baiting Stone	Stealth + Charm	Charm a Bloomling elsewhere to lure them off
4 Maze	C27	Fading Rooms	Reset	Let paint fade to reset a wrong seal pattern
4 Maze	C28	Listener Nest	Stealth + Relay	Moss re-emits loudly - plan the chain, then go quiet
4 Maze	C29	Mirror Wing	Symmetry	Two mirrored rooms share one Hushling
4 Maze	C30	The Long Quiet	Stealth endurance	A full chamber solved almost silently
4 Maze	C31	Whisper Gate	Duet-lite	Low then high while the Listener sleeps
4 Maze	C32	Duskfall	Boss-choir	The maze exhales; the Keep reveals itself
5 Keep	C33	First Chord	Chord	Two thumbs, one door - the new instrument
5 Keep	C34	Answering Stones	Duet	The stones want low, then high, in time
5 Keep	C35	Elder Sleep	Chord + Stealth	The chord wakes the Keep - and its Listener
5 Keep	C36	Choir Stairs	Chord + Relay	One chord, relayed, opens a stair-sequence
5 Keep	C37	The Great Root	All verbs	Grow, charm, shatter, silence - one chamber, everything
5 Keep	C38	Duet of Doors	Duet + Paint-match	Two seals, two voices, one order
5 Keep	C39	Elderchoir Rise	Boss-choir	The choir sings a call; you answer in your own key
5 Keep	C40	Songfall	Finale	The Loom remembers: every biome color returns

Table 7.2 - Complete chamber inventory at launch

7.6 Content volume summary

Content type	Launch quantity	Session role	Post-launch expansion
Song chambers	40 across 5 biomes	Core 3-minute loop; ~60 minutes	+2 chambers monthly, +1 biome per quarter
Daily Loom	1 per day, generated	Daily retention + sharing	Seasonal template packs
Bloom Garden	1 persistent canvas	Creative segment, cosmetics home	Seasonal ornaments
Melodies	32 collectible songs	Collection and Garden play	Event melodies
Instruments / palettes	12 / 8	Self-expression, rewarded economy	Monthly cosmetic drop

Table 7.3 - Launch content inventory

8. Puzzle and Level Design

A song chamber is a small, legible room with one clear goal, one twist, and one celebration. This chapter defines the grammar that every chamber is authored against, the pattern library they draw from, and the difficulty doctrine that keeps the curve gentle without making the game trivial.

8.1 Chamber grammar

- **Goal.** Exactly one visible goal per chamber: the Bloom Door, a charmed Elder, a fully grown bridge to the exit bell. Single-word label, or no label at all when the shape explains itself.
- **Affordance.** The room's first readable object telegraphs the intended verb (sleeping Bloomling beside the path; a gap with a root stump; crystal glinting at the edge of darkness).
- **Aha.** The twist: the moment the room re-frames itself - the moss that relays your note around a corner, the Hushling you can lead away by singing from a ledge. One per chamber, never two.
- **Celebration.** The resolution bloom: the chamber's painted regions animate into a final cascade, the melody resolves to tonic, confetti-leaf particles, Harmony Notes count up. Non-negotiable on every chamber, per the platform's milestone-congratulation doctrine.

8.2 The pattern library

Pattern	Core question	Appears	Example
Reveal	What is in this room?	B1-3	Chirps map the dark; a hidden alcove holds a shard
Grow	Where do roots reach?	B2	Sequence root stumps so a bridge completes before the wave fades
Charm	What does the Bloomling want?	B1-4	Wake it with mid tone; it hums your note into a far switch
Shatter	Which crystal breaks?	B3	One of four crystals dams the water; breaking the wrong one drains the melody
Relay	How does the note travel?	B3+	Echo Moss chain carries color around a wall to a sealed door
Silence	When do I stop?	B4	Sneak the Hushling into a side alcove, then sing the answer freely
Sequence	In which order?	B4+	Duet stones must hear low, then high, or the chord fails
Paint-match	Which colors where?	B3+	Door opens only where floor paint matches its two-tone seal
Duet	Can I hold two voices?	B5	Two-thumb chord; later, one voice plus one Bloomling
Restore	Can I repair it?	B2+	Re-paint a faded pattern to re-awaken dormant moss
Boss-choir	Can I conduct?	B5 finale	The Elderchoir sings a call; answer with the matching phrase

Table 8.1 - Puzzle pattern archetypes (the authoring palette)

8.3 Difficulty doctrine

The curve follows three rules. First, no chamber may require a verb the player has not already been taught in a softer context; introduction chambers for each verb are designed so that the intended note is the only one with any effect, making the lesson self-confirming. Second, difficulty rises by combination and timing, never by precision demands or time pressure - the game never asks for faster fingers, only clearer thinking, which is what keeps the cozy contract intact for the broad audience. Third, there are no difficulty walls: any chamber that loses more than a set share of players in playtest recordings (the exact threshold is in Chapter 15) is either re-taught, re-tuned, or cut. Optional depth - the three-star thresholds, shard hunting, and Daily Loom grading - carries all of the expert-level challenge so the main path can stay generous.

8.4 Teaching without text

Tutorialization is visual and musical only, per the platform's global-audience rule. A new verb is taught in four beats: the object is shown dormant with a strong silhouette; the correct pitch is previewed by the lantern heart's color pulse when the player sweeps near it; singing nearby makes the object visibly stir before it fully reacts (a half-step tell that rewards the first exploratory note); and success triggers the celebration with the

verb's jingle, which becomes its audio signature everywhere else in the game. No popup, no arrow, no paragraph. The entire first session should be understandable to a player who cannot read any language the game ships in - and to one who has muted the game, thanks to the lantern heart's color preview system, which doubles as the accessibility channel (Chapter 16).

8.5 Sample chamber walkthrough

Rootside Glen C5, The Two Stumps. The room opens on a dark gulf with a visible exit bell across the gap; two root stumps sit on near ledges, one thick, one thin; an Echo Moss patch clings to the far wall above the exit. The affordance: stumps are growth objects, so Root Hum is telegraphed. The naive solution fails softly - singing at the thin stump grows a spindly bridge that sags under the Hummy's hops and fades before the crossing completes. The aha: grow the thick stump instead, and while its bridge matures, sing a mid note upward so the moss relays Lantern Tone onto the exit bell's bracket, waking a sleeping Bloomling that holds the door. The player crosses the finished root bridge, hums a soft chord with their new companion, and the chamber resolves in a green-and-amber cascade. Notes used: four. Stars sit at three notes, 55 seconds, and 85% paint coverage - the speedrun line is singing thin stump then thick stump in one held sweep, which top players will discover and share.

9. Game Feel and Juice

Feel is the difference between a game that reads as premium and one that reads as generated. This chapter is the polish contract: every interaction in Songloom has a specified audio, visual, and controller-level response, tuned to the animation principles that the game-feel literature has converged on - anticipation, squash and stretch, follow-through, hitstop, screen shake, particles, escalating celebration. Nothing in the game is allowed to be a dead interaction.

9.1 The juice contract

Interaction	Body / camera	Audio	VFX	Timing rule
Hop	Squash 85% on crouch, stretch 115% in flight, squash 90% on land; camera eases +6px	Soft plucked minor 3rd, pitch rises with hop length	3 dust motes on takeoff, 6 on landing	Anticipation 90 ms before motion
Chirp	Lantern heart pulses 1.2x	Pure sine blip with fast decay	Thin 2.5-tile ring, 4 spark particles	Wave launches on note onset, not release
Held note	Hummy bobs on the beat grid; heart glows in pitch color	Note sustains with slow chorus vibrato	Ember particles orbit the Hummy, density with hold time	Vibrato enters after 600 ms
Bloom wave reveal	Geometry eases in with 120 ms stagger per tile ring	Delay-tail echo of the note; surfaces add faint pad tone	Ink outlines draw themselves; interior fills saturate over 400 ms	Stagger never exceeds wavefront
Verb success (grow)	Root grows joint by joint with overshoot settle	Rising arpeggio in the note key	Leaf-burst at each joint	Overshoot 8%, settle 200 ms
Verb success (shatter)	4-frame hitstop, camera shake 3px, 80 ms	Glassy FM cluster, noise splash	12 shard particles with gravity, glitter trail	Hitstop only on intended breaks
Chamber resolve	Full-screen bloom cascade; camera slow dolly out	Melody resolves to tonic; finale layer swells	Confetti leaves; painted regions shimmer in wave order	2.2 s minimum, never skippable before rewards

Table 9.1 - Interaction response matrix (extract)

9.2 The five-second wow

The first bloom is the product's thesis statement and its player-fit weapon. Before any tutorial beat, the very first input - which is a single tap anywhere - produces the first chirp, and the first chirp must already be beautiful: a warm tone, a ring of light that reveals a sliver of the world, and three particles that catch the light. The first held note, seconds later, rolls the first full wave and colors an entire wall. Session zero's first ten seconds contain the entire game's promise, visible and audible, before a single instruction has been given - which is what conversion-to-play and early playtime are made of.

9.3 Escalating celebration

Celebrations escalate by achievement weight: a charmed Bloomling gets a two-note response and a sparkle; a solved chamber gets the full cascade; a biome finale adds the Elderchoir's answering phrase and a sustained chord that the player's own notes complete. The final layer of the adaptive score - the melodic fragment

bank - only fades in at high resonance, so the most musical players literally hear the world singing along with them. All celebration moments double as the shareable beats in streamer clips: each is designed to read completely in a two-second GIF, with the painted result framing itself.

9.4 Feel QA

- No interaction without response: input to first feedback within 100 ms, always.
- No dead audio: every menu, every state change, every UI affordance has a synthesized micro-cue (Chapter 11 inventory).
- Screen shake is capped: 3px amplitude, 80 ms decay, never during held singing - the player's pitch control must stay stable.
- Hitstop only on intended world events, never on player error, because there are no player errors to punish.
- The juice checklist is a named Claude review prompt (Chapter 23) applied to every new interaction before merge.

9.5 The timing constants

Constant	Value	Why this value
Input to first feedback	under 100 ms	Perceived as immediate; the web audience patience floor
Hop total (anticipation, flight, settle)	400 ms	Weighty enough to feel physical, fast enough to keep pace
Anticipation before action	90 ms	Readable intention without sluggishness
Bloom wave speed	3.2 tiles per second	Wave outruns a hop, so singing leads movement - the strategic order
Chirp wave radius	2.5 tiles / 0.8 s	Quick probe that never over-reveals
Held wave duration	up to 4 s	Long enough to cross a chamber, short enough to stay a decision
Hitstop on shatter	4 frames at 60 fps	Impact without control loss
Screen shake	3 px, 80 ms decay	Felt, never disruptive; pitch drag unaffected
Paint saturation ramp	400 ms to full	Color arrives like light soaking into paper
Celebration minimum	2.2 s, unskippable to payout	The reward beat that retention is built on
Music layer crossfade	1.2 s	Layer changes are felt as mood, not as a cut
Cold load to first note possible	under 5 s	The platform instant-play doctrine, made literal

Table 9.2 - The numbers that govern feel (single source of truth)

10. Art Direction

The visual identity has one job: to make silence look like a promise and sound look like a gift. The world starts as elegant, near-monochrome ink linework - hand-drawn confidence, not emptiness - and singing saturates it into a warm, glowing painting. The art direction is therefore a two-layer system: a static ink world and a dynamic bloom world, with the player's voice as the brush between them.

10.1 The two-layer world

- **Ink layer.** Line-first rendering on warm paper tones: charcoal outlines (#2a2723 on #f5f0e6 paper), soft noise texture, breathing lantern glow around the Hummy. Pre-reveal geometry sits at 8% ink so the dark is readable, never murky.
- **Bloom layer.** Painted surfaces carry the verb's color at 40-90% saturation with a subtle interior glow; paint edges keep the ink outline for cohesion. Global rule: ink is structure, bloom is emotion.
- **The palette.** Five verb colors plus paper and ink: Moss #6d8b5a (low), Amber #c9973b (mid), Sky #5a8fb5 (high), Iris #7d6a9e (chord), and Faded #8e8e8e (silence). All tuned for mutual distinctness

under colorblind checks (Chapter 16).

10.2 Character design language

Every creature is built from two or three shapes and reads perfectly in silhouette at 64 pixels - the thumbnail test doubles as the character test. The Hummy is a rounded capsule body, a heart-shaped lantern chest, and two stub feet: instantly cute, instantly readable, and its lantern is the game's diegetic UI (pitch preview, resonance glow, and mood all live on the chest). Bloomlings are single-circle bodies with petal collars; Hushlings are soft parallelogram sheets that visibly relax when they settle; the Elderchoir is the only large silhouette in the game, reserved for finale weight. Expressions are handled by lantern color and body squash rather than drawn faces, which keeps the cast legible at every zoom and lets the animation system carry emotion cheaply.

10.3 UI philosophy

UI is diegetic and thumb-zone only. There is no HUD clutter: resonance lives on the lantern, notes used live in the world's own ink counter, and the pause surface is a single sheet of paper that slides from the bottom edge. All interactive targets sit in the bottom third of the portrait screen or respond anywhere (tap-to-hop regions cover the whole canvas). Menus are replaced by places: biome maps are hand-inked postcards, and first-time players never see one - they wake up mid-chamber. Typography is a single display family (rounded serif, subset WOFF2, under 60 KB) used only for single-word labels and numbers, honoring the text-light doctrine for a global audience.

10.4 Thumbnail and first-impression design

The thumbnail is a distribution weapon and is designed as such: a 1200x900 canvas with the Hummy mid-song at center-left, a bloom wave rolling to the right in three colors against deep ink, the title in the display serif, and nothing else. It is composed to read at 120 pixels wide on a category page against neighboring thumbnails, where dark canvases with a single bright subject out-earn busy frames. Two alternates ship for web fit test comparison: one zoomed on the Hummy's face, one wide on a painted chamber. First impression in-game is governed by Chapter 9's five-second wow.

10.5 Motion and VFX standards

- Easing: cubic out for all reveals, spring (overshoot 8%) for growth and UI, expo out for celebrations. No linear motion anywhere.
- The bloom wave is the signature VFX: a soft-edged ring with a chromatic fringe in the verb color, leading particles on the wavefront, and ink-draw reveal staggered behind it.
- Particles are quiet and intentional: 3-12 per event, leaf or mote shapes, gravity only where diegetic. No fire, no sparks, no stock explosions - this is not that kind of magic.

- Paint persistence: a global bloom-lighting pass tints the chamber's ink layer toward painted regions at 15% so the room's mood follows the player's song.
- All VFX ship in three quality tiers (particle counts and the lighting pass scale down) for the device matrix in Chapter 13.

10.6 The not-generated style guide

Because the market is flooded with AI-lookalike assets, the style guide is written to make Songloom visibly authored. Ink lines have human variance: stroke width jitters within a fixed band, corners have hand-tuned joins, and every biome's linework carries small deliberate asymmetries (a signature motif per biome - the Hollow's curling fern ticks, the Terraces' stacked drips). Palettes are hand-tuned, never generator output. Animation curves are authored per object family rather than shared globally. And the audio, being synthesized from scratch in code, is by construction un-cloneable from any sample library. The craft budget goes exactly where players look: the first five seconds, the celebrations, and the thumbnail.

11. Audio Design

Audio is Songloom's identity and its moat. The entire soundscape - every note, every jingle, the adaptive score itself - is synthesized in code through the Web Audio API at runtime. The build ships zero audio files: this keeps the package tiny (a hard platform requirement), gives infinite variation without asset cost, and produces a sonic signature that cannot be traced to any sample pack. This chapter is the audio bible: voices, music system, effects inventory, mix, and the mobile constraints that shape all of it.

11.1 The instrument voices

The Hummy's five voices are small synthesis graphs. Root Hum is a triangle-wave oscillator two octaves down with a slow lowpass sweep and gentle vibrato at 4.5 Hz - weight and warmth. Lantern Tone is a sine core with a fifth-above harmonic at minus 12 dB and chorus detune of plus and minus 6 cents - the friendly center of the game's music. Star Chime is FM synthesis with a 2.1 modulator ratio, fast decay, and a noise-transient attack - crystalline and precise. The Chord layer simply routes two of the above into a shared reverb bus with a slight strum delay of 30 ms between voices. Silence is not absence: it is the room tone, a filtered brown-noise bed at minus 38 dB with the delay tails of whatever was just sung, so quiet moments still feel like the same world listening.

11.2 Voice synthesis parameters

Voice	Core oscillators	Filters / FX	Envelope (A/D/S/R)	Character
Root Hum	Triangle, -2 octaves	Lowpass 400 Hz sweeping to 900; vibrato 4.5 Hz, 12 cents	120 ms / 300 ms / 0.7 / 900 ms	Warm, woody, patient
Lantern Tone	Sine + fifth harmonic -12 dB	Chorus detune +/-6 cents; plate reverb send 20%	80 ms / 200 ms / 0.8 / 700 ms	Friendly, round, central
Star Chime	FM pair, carrier 1568 Hz, ratio 2.1	Highpass 800 Hz; shimmer +5 cents; noise transient 15 ms	5 ms / 400 ms / 0 / n/a	Crystalline, precise, bright
Chord	Two of the above, strum 30 ms apart	Shared reverb bus; gentle compressor 2:1	per component voice	Ceremonial, resolving
Silence	Filtered brown noise bed	Lowpass 300 Hz; previous delay tails at 30%	n/a (ambient)	Attentive, breathing room

Table 11.1 - The five voices: synthesis graph parameters

11.3 The adaptive score - vertical layering

The music system uses vertical re-orchestration over a fixed harmonic grid. Every biome has a chord loop of eight bars at 76 BPM in a major or dorian tonality, and four layers stack on it: the base drone (always present, low-passed pad), the biome layer (a soft motif - the Hollow's owl-like vowel pads, the Terraces' glass droplets), the activity layer (percussive leaf textures and rhythmic plucks that fade in with player action density, so movement itself raises the music), and the melodic fragment bank (short call-and-response phrases that only enter at high resonance, harmonizing with the key the player has been singing). Layers crossfade over 1.2 seconds with gain automation; the whole system is six parallel synthesis chains and a scheduler, roughly 600 lines of TypeScript, and it makes every session sound slightly like a small ensemble improvising around the player.

11.4 The pitch-verb feedback loop

The player's own notes are always the loudest musical element, and they are quantized softly to the current key: singing near a chord tone snaps gently to it, so even improvised wandering sounds musical, and the world's verb response fires on the quantized pitch. This single decision - the game secretly wants to sound good - is what makes the core loop feel like performance rather than input. Bloom waves carry the note through a tempo-synced delay (dotted eighth at 76 BPM) whose feedback level rises with resonance: late in a good chamber, the room is audibly full of the player's own echoing melody. This is the game's signature sound and its most shareable property.

11.5 Effects inventory (extract)

Event	Recipe (synthesis)	Level	Notes
Hop landing	Filtered noise burst 40 ms + low triangle thump	-20 dB	Pitch scales with hop distance
Chirp	Sine 900-1400 Hz, 90 ms, fast decay	-12 dB	Wave onset sound
Bloom reveal	Note copy through delay + shimmer (pitch-shift 5 cents)	-18 dB	The echo tail
Grow joint	Rising triangle gliss 200 ms, pluck transient	-16 dB	Arpeggio up the root chain
Shatter	FM cluster (ratios 2.1, 3.4) + noise splash 120 ms	-14 dB	Hitstop partner
Bloomling wake	Two-note vowel pad, major third, 400 ms	-16 dB	Answers in the player key
Hushling settle	Downward filtered noise sigh 600 ms	-24 dB	Relief cue
Chamber resolve	Tonic triad swell + finale layer entry	-10 dB	2.2 s minimum
Harmony Notes tick	Soft wood block, pentatonic ascent	-22 dB	Count-up rhythm
UI tap	Muted pluck 60 ms	-26 dB	Every menu sound is a note in key

Table 11.2 - Sound effects inventory and synthesis recipes (extract of 60+ total)

11.6 Mix architecture

The mix runs on four buses with a master limiter at minus 1 dB true peak: the Music bus (score layers, minus 20 LUFS), the Voice bus (the player's notes, minus 16 LUFS and side-chain priority), the World bus (verb events and creatures, minus 18 LUFS), and the UI bus (minus 26 LUFS). Side-chain ducking is gentle and one-directional: the world ducks 2 dB under the player's voice, never the reverse - the player's instrument is the soloist, always. A loudness cap of minus 14 LUFS integrated keeps the whole game comfortable for long sessions, which suits the cozy register and avoids the fatigue common in ad-supported web games where audio over-compression reads as cheap.

11.7 Mobile audio constraints and accessibility

Two platform realities shape the engine. First, audio contexts on iOS and some Android browsers must be unlocked by a user gesture, so the first tap of the loading screen both starts the game and unlocks the audio graph, with the first chirp doubling as the unlock confirmation - the constraint becomes an onboarding beat. Second, output latency varies from negligible on desktop to tens of milliseconds on mid-tier Android, so all rhythm-adjacent logic (hop animation grids, delay synchronization) reads a measured latency constant at startup rather than assuming a fixed value. Accessibility: the lantern heart's color preview mirrors every pitch cue visually, the wave itself is the visual echo of every sound, and a captions-lite mode adds named ripples for key creature events - the game is fully playable muted by design, which also covers the

silent-browsing contexts web players actually sit in.

12. Technical Architecture

The stack is chosen for the platform's constraints first and the developer's strengths second: TypeScript compiled to a single bundle, PixiJS v8 as the WebGL2 renderer with custom fragment shaders for the bloom waves, the Web Audio API for everything audible, and no full game engine. The reasoning is simple: engine runtimes (Unity, Godot, Phaser-as-framework) carry hundreds of kilobytes to megabytes of capabilities Songloom does not use, and the platform's file-size guidance is a hard distribution advantage. A framework plus 100% procedural content keeps the initial download at roughly three megabytes with zero compromise on visual ambition, because the game's entire look is shaders, ink geometry, and palettes rather than heavy assets.

12.1 System map

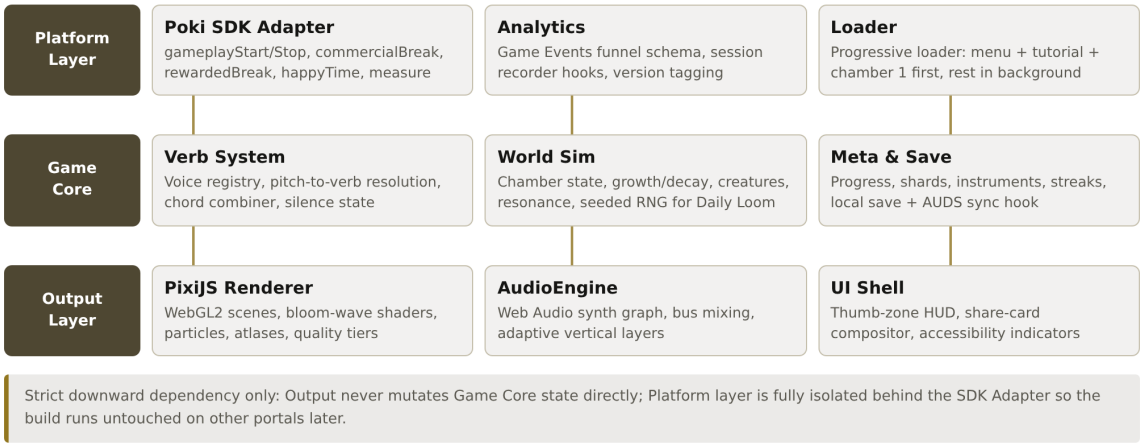


Figure 12.1 - System architecture: platform, game core, and output layers with strict downward dependencies.

Module	Owns	Approx. size	Notes
Core loop / state	Chamber state machine, input routing, save	40 KB	Deterministic update at 60 Hz fixed step
Verb system	Voice registry, pitch-to-verb, chord combiner	25 KB	Data-driven; verb params in chamber JSON
World sim	Growth/decay, creatures, resonance, seeded RNG	70 KB	Chambers are data, not code
Renderer	PixiJS scenes, bloom shader, particles, tiers	120 KB	Custom shader for wave + ink draw
AudioEngine	Synth graphs, bus mixing, adaptive scheduler	80 KB	Zero audio assets, ever
UI shell	Thumb HUD, share-card compositor, captions-lite	60 KB	Canvas-composited share images
Poki adapter	SDK wrapper + event schema	15 KB	Isolated; other portals later
Loader	Progressive stages, progress events	20 KB	Stage 1 bundle <1.5 MB

Table 12.1 - Module responsibilities and size budgets

12.2 The chamber data format

Every chamber is a JSON document validated against a schema at build time, which is what makes a solo-plus-AI production line possible: Claude generates candidate chambers from the schema and the pattern library, the validator rejects anything unsolvable or off-doctrine, and the human gate is play and taste. A chamber declares its anchor graph (hop points and arcs), verb zones (regions with verb responses and parameters), creature spawns, the goal condition, star thresholds, event tags for analytics, and the palette variant. The Daily Loom generator emits the same schema from a constrained template grammar, with a deterministic seed (YYYYMMDD integer) so one date equals one chamber worldwide. Sample documents are in Appendix B.

Schema extract

```
chamber.schema.json (extract)
{
  "id": "rs_glen_c05",
  "biome": 2,
  "size": [14, 20],
  "anchors": [ {"id": "stump_a", "x": 3, "y": 11, "type": "growth_thick"} ],
  "verbZones": [ {"zone": "gap_main", "verb": "grow", "target": "bridge_a",
"maturity": 2.4} ],
  "creatures": [ {"type": "bloomling", "at": "bell_bracket", "carries": "mid"} ],
  "goal": { "type": "bloom_door", "requires": ["bridge_a.complete",
"bloomling.charmed"] },
  "stars": { "notes": 3, "time_s": 55, "paint_pct": 85 },
  "events": ["level", "rs_glen_c05", "start|complete|fail"]
}
```

12.3 Determinism and the fixed-step core

The simulation updates at a fixed 60 Hz with an accumulator, rendering decoupled and interpolated, so physics-adjacent behavior (wavefronts, growth timings, creature drift) is identical across frame rates and devices. All gameplay randomness flows from a seeded PRNG (mulberry32), and the Daily Loom seed is derived from the date alone. This determinism is also the QA foundation: any bug report reproduces exactly from its seed and input log, and the smoke test suite replays recorded inputs against golden hashes of the chamber state after solve.

12.4 Code quality gates

- TypeScript strict mode, zero any at merge; public module surfaces documented with TSDoc comments.
- Unit tests for the verb system and world sim (target: 85% line coverage on those two modules).
- Schema validation on all chamber JSON at build time - invalid chambers fail CI, not playtest.
- Bundle size regression gate: stage-1 bundle over 1.6 MB or total over 6 MB fails the build.
- No runtime dependencies beyond PixiJS and the Poki SDK; everything else is first-party code.

13. Performance and Loading

Performance is a distribution feature on the web, not an engineering nicety. The platform's own numbers frame the budget: initial download at or under five megabytes, total at or under eight, and drop-off during loading described as brutally underestimated. Songloom targets roughly three megabytes initial and under six total, first playable in under five seconds on a mid-tier Android phone over a average connection, and a stable frame rate at 60 on desktop and 30-plus on the reference low-end device. Every technique below exists to serve one of those three numbers.

13.1 The loading plan

- Stage 1 (target under 1.5 MB, first playable): loader shell, core loop, verb system, ink renderer, audio engine, tutorial chamber. Players are singing within seconds.
- Stage 2 (background): biomes 1-2 chamber data, the Daily Loom template grammar, UI shell and share compositor, the display font subset.
- Stage 3 (background, cached): remaining chambers, garden systems, cosmetic definitions. IndexedDB caching with a version key so repeat visits load in under a second.
- The loading screen is a hand-inked progress card with the Hummy breathing and a single word - LOOMING - while a progress ring fills from real byte events. No spinner, ever: spinners read as waiting, and waiting is the enemy.

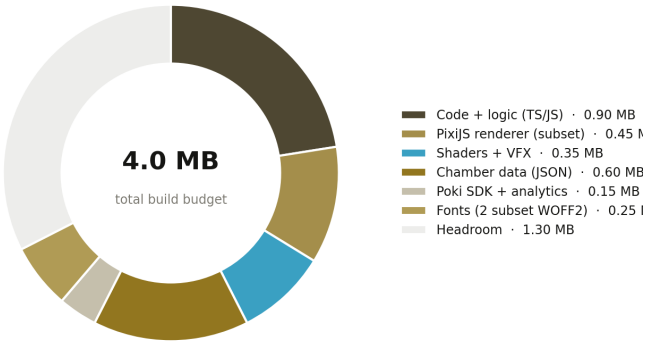


Figure 13.1 - Total build budget allocation (4.0 MB target against the platform 8 MB ceiling).

13.2 Frame rate discipline

The renderer is organized around three costs. Draw calls: the entire ink layer is one static batched mesh per chamber plus a second dynamic batch for paint regions; creatures and particles live in shared sprite atlases (two 512-pixel atlases hold the whole game's art). Fill rate: the bloom wave is a single full-screen-equivalent shader pass with an early depth reject against the ink mask, so dark regions cost almost nothing; the wave shader itself is 40 lines of GLSL with no branching per pixel. Script: fixed-step sim under 2 ms per tick on the reference device, enforced by a CPU budget assert in the debug build. Particles cap at 220 alive with pooling - no allocation in the steady state - and the GC sees zero churn per frame by design, because frame hitches on low-end Android are visible as stuttered waves, and a stuttered wave is a broken promise.

13.3 Device tiers

Tier	Reference devices	Profile	Key differences
T1 high	Desktop, flagship phones	Ultra	Full particles, bloom lighting pass, 2x supersample on waves
T2 mid	iPhone SE class, mid Android	High	Full particles, lighting pass at half resolution
T3 low	2-3 GB RAM Android	Standard	Particle budget halved, lighting pass off, wave shader simplified
Fallback	No WebGL2	Compatibility	Canvas2D ink mode, waves as pre-drawn rings, audio unaffected - protects the long tail the platform explicitly serves

Table 13.1 - Quality profiles by device tier

Tiering is automatic from a one-second benchmark probe at boot (fill-rate and script timing samples), never from user-agent sniffing, and the tier is reported as a game event so analytics can correlate performance with playtime. WebGPU is treated exactly as the platform's principal engineer advises: where available, the wave pass runs on a compute-friendly path with batched command submission; where not, WebGL2 carries the game identically. No feature is WebGPU-only, because thirty-two percent of the platform's players would be locked out of it.

13.4 Optimization techniques register

- Procedural everything: audio synthesized, palettes computed, particles parameterized; the only binary assets are two texture atlases and a subset font.
- Texture atlases at 512 px with premultiplied alpha; sprite frames share power-of-two padding.
- Chamber data as flat JSON with integer grids, parsed once into typed arrays.
- Object pooling for waves, particles, and audio voices; steady-state allocation zero.
- Code splitting into the three load stages; the share compositor and garden code never load during a first session.
- Audio voices are destroyed and rebuilt per note rather than kept alive - Web Audio node graphs are cheap to construct and expensive to accumulate.

14. Poki SDK Integration

The SDK integration is specified as carefully as the game itself, because two of the platform's measured metrics depend on it: conversion to play is defined by the first gameplayStart event, and every monetization surface flows through commercialBreak and rewardedBreak. The adapter module isolates the SDK behind a six-method interface so the same build runs on other portals later with a shim, and every call site is listed here so nothing is left to memory during implementation.

14.1 Event calls

Integration reference

```
SDK call sites (adapter interface)
// Boot: init, then load gameplay assets
await PokiSDK.init();                          // once, before gameplay

// Session lifecycle
PokiSDK.gameplayStart();                      // first player input of the session
(C2P!)
PokiSDK.gameplayStop();                       // pause menu open, tab hidden
(visibilitychange)

// Natural pauses - never interrupt a held note
if (chamberResolved) PokiSDK.commercialBreak();   // between chambers,
post-celebration

// Player-initiated only
const rewarded = await PokiSDK.rewardedBreak();   // hint, streak repair, preview
flows
if (rewarded) grantReward(placement);

// Celebration moment
PokiSDK.happyTime();                           // chamber resolve, streak milestone

// Analytics funnel
PokiSDK.measure('level', chamberId, 'start');
PokiSDK.measure('level', chamberId, 'complete');
PokiSDK.measure('rewarded', 'echo-hint', 'visible');
PokiSDK.measure('rewarded', 'echo-hint', 'interact');
```

14.2 Placement rules

- gameplayStart fires on the first meaningful input of the session - the loading screen's first tap, which is also the audio unlock. This guarantees C2P measurement begins exactly when interactivity begins.
- gameplayStop fires on pause, on tab-hidden, and on session end; commercialBreak plays only at natural pause points: between chambers after the celebration completes, and on the biome map. Never mid-note, never mid-wave, never during a celebration.

- rewardedBreak is player-initiated only, always paired with a visible non-rewarded alternative in the primary position, per the platform's monetization doctrine (Chapter 17).
- happyTime fires at chamber resolve and streak milestones - the game's genuine delight moments, which is the call's intended emotional slot.
- measure events follow the drop-off funnel schema: level start / complete / fail per chamber, rewarded visible / interact per placement, shop visible / interact for the garden and instrument surfaces.

14.3 Loading contract and common bugs

The loader reports real progress to the SDK so the platform can display it during playt loading, and the first gameplayStart only fires when the player can actually act. The three integration bugs that most commonly damage web fit test results are all designed out here: a missing gameplayStart (C2P reads as zero - guarded by the input-gated call above), commercial breaks firing during interaction (reads as interruption and depresses playtime - guarded by the natural-pause rule), and unlocked audio state on iOS causing a silent game (the first tap owns both unlock and gameplayStart, and the adapter retries init once if the audio context starts suspended). The Poki Inspector is run against every release candidate as a pre-submission gate, and its warnings are treated as release blockers, not advice.

15. Analytics and Data

Analytics exists to answer exactly one class of question: where does the game lose players, and what should change because of it. Everything measurable is an event through the platform's measure schema plus game-side session metadata, and every metric below has a named target and a named action when the target is missed - a dashboard without a decision rule is decoration, not development.

15.1 Event taxonomy

- Session events: start, tier, language, chamber reached, notes sung, paint coverage, duration.
- Funnel events per the platform schema: level start / complete / fail, chamber by chamber.
- Verb events: which voice was used where - validates the teaching order and surfaces confusion (sweeping without releasing, mid-holds with no verb target).
- Rewarded events: visible / interact per placement; reward claimed; streak repaired.
- Share events: song card generated, shared; daily loom entered, completed, graded.

15.2 Metric targets and decision rules

Metric	Target	Measured by	Action if below target
Load to first input	under 5 s on T3	Session events, tier-tagged	Cut stage-1 bundle; move font to stage 2
First-session chambers	4+ completed	Funnel, new players	Soften C1-C3; move first surprise earlier
Player fit test average	4+ minutes	Poki player fit test	Rework mid-biome chambers flagged by recordings
Chamber 1-4 completion	95%+ each	Funnel	That chamber is broken; re-teach or cut
Any chamber completion	below 60%	Funnel	Playtest recordings, then re-tune or cut
Daily Loom return rate	18%+ of active players	Share + daily events	Streak reward tuning; share card redesign
Rewarded interact rate	8-15% of visible	Rewarded events	Reposition placement; never increase pressure

Table 15.1 - Core metrics, targets, and pre-committed actions

15.3 The iteration loop

The loop that turns this data into the game is weekly and deliberately boring: pull the funnel, watch playtest recordings for the two worst chambers, write a fix hypothesis with a success criterion, ship it, and let the next player fit test adjudicate. Claude's role in this loop is triage - timestamped recording notes in, prioritized hypothesis list out - and the developer's role is judgment. The playtest cadence (Chapter 22) aligns to the two-tests-per-day player fit limit so every build is measured before the next one exists. The discipline is the point: a solo developer's scarcest resource is attention, and this loop spends it only where the data says players actually struggle.

16. Quality Assurance and Testing

QA for a solo production cannot be a department; it has to be automation plus platform playtests plus a checklist culture. This chapter defines the device matrix, the automated suites, the manual passes, and the accessibility gates that every release candidate must clear before it is uploaded to the platform.

16.1 Device matrix

Class	Device (representative)	Browser	What it proves
Low Android	2 GB RAM, 2019-era chip	Chrome	T3 profile holds 30+ fps; load under 5 s on slow 3G emulation
Mid Android	4 GB, 2021-era chip	Chrome	T2 default tier picks correctly; audio latency path
iPhone SE	current SE class	Safari	Audio unlock flow; safe-area insets; portrait lock
iPad	standard iPad	Safari	Landscape letterboxing correct; touch targets generous
Desktop	any 2015+ laptop	Chrome + Firefox	60 fps; keyboard parity; pointer-as-touch parity
Fallback	no WebGL2 (rare)	Chrome	Compatibility ink mode is fully playable

Table 16.1 - Reference device matrix (the floor for sign-off)

16.2 Automated suites

- Smoke suite: headless Chromium replays 12 recorded input scripts (one per pattern archetype) against every release candidate; chamber state hashes must match goldens.
- Perf suite: scripted 60-second sessions on throttled CPU and network profiles in CI; assert frame-time percentiles and stage timings against budget.
- Schema suite: all chamber JSON, including generated dailies for the next 90 days, must validate and prove solvable by the solver pass.
- Audio suite: offline render of the synth graph for every event recipe; null-test against reference renders to catch regressions from parameter drift.
- Save suite: save/load round-trip fuzzing; version migration from every prior public schema.

16.3 Accessibility gates

Accessibility is a distribution requirement as much as an ethical one, because the platform's audience is global, multi-device, and frequently muted. The verb palette is verified for deuteranopia and protanopia distinctness (Moss and Amber shift toward luminance separation in a colorblind-safe variant that swaps hues while keeping ink contrast); all pitch cues have the lantern heart color mirror; the captions-lite mode adds named ripples for creature events; the UI has no audio-only or color-only information anywhere; and the whole game is operable with one finger or one keyboard hand. The muted-playthrough is part of the sign-off: a player who never hears the game must still be able to finish biome one without confusion, because the wave is the visual echo of every sound the game makes.

16.4 Release candidate checklist

- Poki Inspector clean (no warnings) on the candidate build.

- Smoke, perf, schema, audio, save suites green in CI.
- Device matrix signed off manually, screenshots archived.
- Muted playthrough of biome 1 complete without confusion.
- Event verification: gameplayStart, break, rewarded, happyTime, measure funnel all observed in a staging pass.
- Loading screen progress reflects real bytes; stage 1 under 1.5 MB verified on the built bundle.

17. Monetization Design

Monetization is designed into the game from the first chamber, per the platform doctrine that ads designed in early feel like a favor to the player while ads bolted on at the end feel forced. The model is the platform's own: advertising revenue share on a free game, expressed through a single soft currency and rewarded video placements that respect the cozy contract. There are no hard paywalls, no secondary currencies, and no purchase pressure anywhere in the design - the game is free, warm, and quiet, and the monetization is shaped to stay that way.

17.1 The economy

Harmony Notes are the single currency, earned from chamber completion, star thresholds, resonance bonuses, Daily Loom streaks, and garden milestones. They are spent on garden ornaments, instrument cosmetics, and palette variants. The economy is intentionally shallow and legible: a new instrument costs about six chambers of comfortable play, and every price is visible in one screen with no conversion tables. The platform's own guidance - one understandable currency, clear rewards, no secondary currencies - is adopted verbatim because it is both doctrine and good cozy design.

17.2 Rewarded placements

Placement	Trigger	Reward	Free alternative (primary position)
Echo Hint	Chamber attempted twice without resolve	A shimmer outlines the next verb zone	Explore: the world always reveals enough with patience
Chamber Skip	Chamber attempted three times	Skip with completion credited, stars forfeited	Any chamber is completable; no content is gated
Streak Repair	Daily Loom streak broken	Restore yesterday streak	Accept the loss; streak insurance also purchasable with Notes
Instrument Preview	Cosmetic screen visit	60-second full-timbre preview	Standard preview tone is always audible
Garden Boost	Garden session over 3 minutes	One rare ornament seed	All ornaments reachable through ordinary play
Daily Re-Run	Daily Loom finished, stars lost	One graded re-attempt	Tomorrow's chamber is the fresh start

Table 17.1 - Rewarded video placements, triggers, and alternatives

- Every placement shows the non-rewarded path in the primary position, exactly as the platform's best-practice guide prescribes.
- Rewards are celebrated - animation, sound, automatic application - so the exchange feels like a gift, not a transaction.
- Unlimited rewarded videos are allowed by the platform, but placements here are capped by natural triggers so the economy and streak balance stay protected.
- Dynamic placements (appearing contextually at exactly the moment of sympathetic need) are used instead of persistent buttons, per the platform's own preference.

17.3 Seasonal and portrait revenue

Weekly and seasonal unlocks lift engagement and retention per the platform's monetization tips, so the cosmetic calendar (Chapter 20) is the monetization calendar: one instrument and one palette variant drop monthly, seasonal garden ornaments arrive with each event, and the Daily Loom's seasonal template packs refresh the daily loop's texture. Portrait support is the second revenue lever - portrait-playable games see more gameplay entries and become eligible for the Gamebar Display ad format, which earns passively with no developer work. Songloom is designed portrait-first specifically to bank both effects. Commercial breaks play only at natural pause points between chambers; there are no mid-interaction interruptions anywhere in the design, because interruption is the fastest way to teach a cozy audience to leave.

18. The Poki Submission Kit

This chapter is the operational playbook for getting the game accepted: a self-audit against every published platform criterion, the pipeline walkthrough with dates and deliverables, the category strategy, and the response plan for each possible outcome. The entire chapter exists because the brief's hard requirement is acceptance, and acceptance is earned before submission, not requested during it.

18.1 Pre-submission self-audit

Criterion (official)	Songloom evidence	Where in this PRD
Quality - UX feel and core loop	Single-verb loop with 60+ juice contracts; 5-second wow; full response matrix	Ch. 6, 9
Player fit - audience match	Cozy creativity segment named in platform audience data; no-fail doctrine	Ch. 3, 6
Tech - load, stability, size	3 MB initial / 6 MB total; 5 s to playable; 30+ fps floor on T3	Ch. 13
Originality - fresh mechanic	Pitch-as-verb painting system; genre teardown shows no web or cozy or verb-based precedents	Ch. 4, 5
Depth and replayability	40 chambers, 3-star mastery, daily loop, garden, collection; loop opens over time	Ch. 7, 8
First impression	Thumbnail kit; menu-less onboarding; first note blooms within 5 s	Ch. 9, 10
Cross-device feel	One-thumb portrait-first controls; keyboard parity; device matrix	Ch. 6, 16
Room to grow	Live-ops calendar and roadmap shared at submission	Ch. 20

Table 18.1 - Self-audit against Poki criteria (the evidence dossier)

18.2 Category strategy

The web fit test allows up to four suggested categories, and the choice shapes which benchmark averages the game is scored against. The strategy is to claim the categories where the game's aesthetics and session shape match the audience, and to avoid the saturated shelves where it would be compared against entrenched incumbents: Puzzle is the primary (session shape, difficulty doctrine, audience fit), Music is second (identity, share artifact, though benchmarked lightly), Skill is third (mastery stars, speedrun surface), and Creative rather than Action is the fourth - Creative pulls the dress-up and coloring audience that the platform itself says responds to self-expression, and it is exactly the segment Bloom Garden serves. Action, io, and 2-Player are deliberately avoided: those benchmarks belong to games Songloom should not be

measured against.

18.3 The pipeline walkthrough

- Month 6, week 1: candidate build passes the release candidate checklist (Ch. 16) and Poki Inspector; upload with thumbnail A and the roadmap one-pager attached to the submission.
- Weeks 1-2: run two playtesting rounds (20 recordings); fix what the recordings show before spending fit tests.
- Weeks 2-4: player fit tests, up to two per day, iterating until average playtime holds at 4+ minutes with 30%+ of plays over 3 minutes, with margin above the 3-minute minimum.
- Weeks 4-8: web fit test (3-5 days per run); iterate on CTR and time-on-page signals - CTR problems point at the thumbnail, C2P problems at load and onboarding. Target: C2P 65%+ and scores of 3+ against category benchmarks.
- On targets met: request final review (1-2 weeks decision). A pass opens the agreement, legal, QA, soft release, and global release sequence, roughly two to three months.

18.4 If the answer is no

A rejection is treated as data with a response protocol, not a dead end. The protocol: request the specific feedback; map every note to a section of this document; rebuild the weakest criterion rather than arguing; and resubmit a meaningfully changed build rather than a renamed one, because the platform's reviewers explicitly say they stay in touch hoping developers return with new concepts. The parallel plan is already in motion by that point - a secondary submission to CrazyGames with the same build - so the work is never single-threaded on one platform's decision. What the plan never does is resubmit the same game unchanged, which is the one move that converts a no into a reputation.

19. The Virality Engine

Virality is engineered as a loop, not hoped for as a lottery. The engine has three parts: the Daily Loom, which gives the whole world one shared puzzle per day; the Song Card, a spoiler-free share artifact that shows the player's journey without leaking the solution; and designed clip moments, which make the game watchable by people who have never played it. The pattern is the one Wordle validated: a common daily challenge, an identity-bearing result card, and curiosity as the only marketing spend.

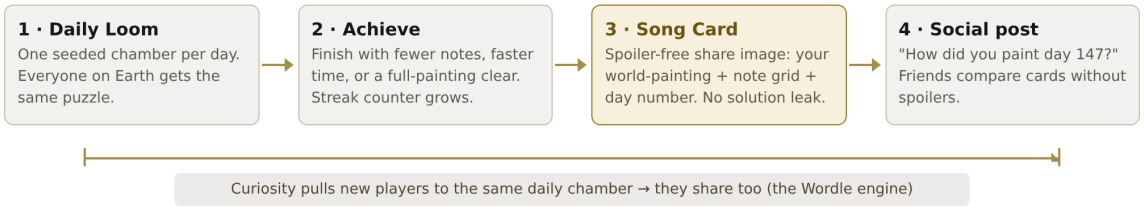


Figure 19.1 - The daily share loop: same puzzle worldwide, spoiler-free cards, curiosity as the conversion engine.

19.1 The Song Card

The share artifact is a rendered image of the player's actual painted chamber - their melody's footprint, not a generic score screen. The card carries the game wordmark, the day number, the note grid (which pitches were sung, as five colored dots per row - legible to players, meaningless to newcomers), stars earned, and the streak flame. It never shows the chamber layout or the solution path, which is the spoiler-free discipline the Wordle grid taught: the share shows the journey, not the answer. Two formats render locally on canvas: square for feeds and 9:16 for stories, both composited from the session's own paint map so no two cards look alike. The platform is web-native, so sharing is a long-press save plus the OS share sheet - frictionless by default.

Field	Content	Source	Rule
Painting	The player's actual painted chamber, top-down, ink layer + bloom colors	Session paint map render	Always the session's own image; never a stock frame
Note grid	Rows of 5 dots, one row per note sung, colored by voice	Input log	Shows the melody shape; no spatial info
Day + stars	Day number, 3-star icons, grade word	Daily result	One-line identity: "Day 147 - Full Bloom"
Streak flame	Streak count with flame icon tiering at 7 / 30 / 100	Streak store	Status worth sharing, not worth boasting
Wordmark	SONGLOOM in display serif + "play at poki.com/songloom"	Static	Legible at 40% scale in a feed
Formats	1080x1080 feed, 1080x1920 story	Canvas compositor	Rendered locally, no network, no upload
Offer timing	Once, at the grade screen	UI flow	Never re-nagged; share is always the player's idea

Table 19.1 - Song Card specification

19.2 Clip moments

- First bloom: the 5-second wow doubles as the 5-second hook - every new player's first note is a potential ad for the game.
- Full-painting clears: a chamber resolving in one continuous cascade reads as a perfect two-second GIF; the star thresholds are tuned so skilled players produce them regularly.
- Secret duets: rare Elderchoir responses to well-sung phrases - discoverable, timestampable, and postable.
- Daily near-misses: the one-note-over star threshold card, which is the exact emotional shape that made Wordle grids compelling - a story of almost.
- Garden showcases: weekly community showcases of player gardens, seeded from the hashtag, giving creative players their stage.

19.3 Where the loops land

The distribution targets are chosen for format fit rather than reach vanity: short-video platforms where satisfying loops and ASMR-adjacent sound design thrive, cozy and puzzle gaming communities, and streamers whose audiences match the register - calm puzzle content rather than rage content. The game's audio identity is itself a distribution asset: a fully synthesized, echoing, gentle soundscape is exactly the kind of thing short-video algorithms already favor in satisfying-content niches. The outreach plan (Chapter 21) seeds the loop with twenty hand-picked creators at launch rather than mass mailings, because one genuine cozy-gaming clip outperforms a hundred cold emails.

19.4 The contract with players

The engine obeys one rule above all metrics: sharing is always the player's idea. No share gates, no viral unlocks, no dark patterns - the card is offered once at the daily's grade screen and never nagged again. This is partly ethics and partly strategy: the cozy audience punishes pressure, and the platform's own anti-pushiness doctrine applies to social prompts as much as ad prompts. The loop's fuel is genuine delight at the daily's result; the design's job is simply to make that delight look good in a card.

20. Live Operations: Year One

The platform is explicit that games which keep improving keep earning promotion, and this calendar is the standing answer to that rule. It is deliberately paced for one developer with an AI pipeline: a monthly rhythm that ships visible value without heroic crunches, and four seasonal tentpoles that give the community something to gather around. The whole calendar is included in the submission package as the roadmap the platform's reviewers ask for - room to grow, demonstrated.

20.1 The monthly rhythm

Slot	Content	Effort source	Purpose
Chambers	+2 song chambers (pattern-library generations, hand-tuned)	Claude draft + human gate	Fresh puzzle content; feed the promoted-update cycle
Instrument	+1 cosmetic voice timbre	Synth parameter work	Economy and self-expressio n refresh
Palette	Bi-monthly bloom palette variant	Hand-tuned color work	Collection and seasonal feel
Daily pack	Seasonal template pack for the Daily Loom	Grammar constraints	Keeps the daily loop textured
Garden	Seasonal ornament set	Ink art, pooled with events	Creative segment retention

Table 20.1 - Monthly live-ops cadence

20.2 The four tentpoles

- Songlight Festival (month 2): the first seasonal - lantern-lit chambers at dusk palettes, a festival garden, and the first event melody; the announcement that the game is alive.
- Echoes of Spring (month 5): growth-themed chambers and a community planting event where gardens visibly link into one shared meadow wall.
- The Great Duet (month 8): the community event - the Elderchoir learns a phrase from the week's most-shared Song Cards and sings it back in a global finale chamber.
- Winter Choir (month 11): the year-one celebration; snow-ink palettes, the choir finale reprise, and the year-two roadmap reveal.

20.3 The year-one calendar

Month	Chambers	Cosmetic	Daily pack	Tentpole / event
Launch	40 live	2 instruments, 4 palettes	Core grammar	Global release week
M1	+2 (Hollow depths)	Instrument: Reed	n/a	First weekly Garden showcase
M2	+2 (Rootside)	Instrument: Bell	Autumn pack	Songlight Festival (tentpole)
M3	+2 (Terraces)	Palette: Dusk	Autumn pack	Community melody vote #1
M4	+2 (Maze)	Instrument: Glass	Winter pack	Great Duet teaser: choir learns
M5	+2 (Keep)	Palette: Frost	Winter pack	Echoes of Spring (tentpole)
M6	+2 (festival set)	Instrument: Wood	Spring pack	Half-year recap + Songbook vol. 2
M7	+2 (Spring glen)	Instrument: Choir	Spring pack	Community melody vote #2
M8	+2 (Duet wing)	Palette: Meadow	Summer pack	The Great Duet (tentpole)
M9	+2 (Summer tide)	Instrument: Shell	Summer pack	Speedrun week: star chase
M10	+2 (Lantern walk)	Instrument: Ember	Autumn pack	Lantern community garden wall
M11	+2 (Choir rise)	Palette: Aurora	Winter pack	Winter Choir (tentpole)
M12	+2 (Year door)	Instrument: Yearbell	Anniversary pack	Year-two roadmap reveal

Table 20.2 - Month-by-month live-ops calendar (year one)

20.4 Operating discipline

Each update follows the same production line as launch content - schema-validated, smoke-tested, playtested with two rounds of recordings before release - so quality does not decay with distance from launch. Every content drop is announced to the platform contact first, per the platform's guidance that timing major drops with promotion multiplies their reach. The dev-time budget for the monthly rhythm is bounded at roughly eight working days per month, which the Chapter 23 pipeline supports indefinitely; anything beyond that scope is deferred rather than crunched, because the game's register and the developer's sustainability are the same asset. Analytics from Chapter 15 arbitrate what the second year doubles down on: if the daily loop retains best, the daily gets the tentpole budget; if the garden community blooms, the garden becomes the heart of year two.

21. Marketing and Community

The marketing plan spends zero cash and respects one constraint: the developer's time is the game's development budget, so marketing must be an exhaust of production rather than a competitor with it. The strategy is build-in-public with a specific hook - a sound engineer synthesizing an entire game world's audio in code is a genuinely rare content angle in a sea of generic AI game-dev videos - plus the platform itself as the primary distribution engine, and a small, warmly-run community rather than a large, coldly-run one.

21.1 Pre-launch (months 4-6)

- Devlog series: 6 episodes on the audio engine, the bloom shader, the daily generator, the chamber pipeline, the juice pass, and the submission story; each ends with a 15-second playable or watchable hook.
- Short-video seeds: 60-second build clips cut from real development footage - the first-bloom moment, wave shader iterations, synth parameter tweaking with audible results.
- Press and creator kit: fact sheet, 12-GIF bank, 3 screenshots, the verb-system one-pager, and a playable demo link - ready before launch week, not after.
- Community seed: a Discord with four channels (daily-talk, garden-showcase, devlog, help) and a weekly prompt seeded by the devlog cadence.

21.2 Launch and beyond

Launch week concentrates everything: the devlog finale publishes as the game goes live, the creator kit goes to twenty hand-picked cozy and puzzle channels with a personal note about the game's angle, and the first Daily Loom runs on day one so the share loop starts spinning with the first players. Post-launch, the cadence is fixed and cheap: one devlog-adjacent post per content drop, one community showcase per week, and daily presence in the Discord for the first month - presence, not performance. The thumbnail and category page performance are the real marketing surface, and they are already owned by the Chapter 18 playbook; social exists to feed curiosity into the platform's own funnel, not to substitute for it.

21.3 The reputation layer

In a market visibly flooding with generated sameness, the developer's visible craft is a marketing asset: responding to player bug reports within a day during soft release, crediting community melody ideas in the Songbook, and publishing the postmortem after launch - these build the reputation that makes the second game's submission pre-trusted. The platform's reviewers say they stay in touch with developers they like; players do the same thing at scale. The plan treats reputation as compounding capital: slow to build, impossible to fake, and the only moat that survives a flood of clones.

22. Development Roadmap: Six Months

The roadmap converts this document into sequenced work with exit criteria, sized for one developer plus the Claude pipeline at a sustainable pace (target 5-6 focused days per week; the timeline absorbs slack in month five rather than pretending it does not exist). Every phase ends with a playable build, because the entire plan depends on real player data arriving early and often.

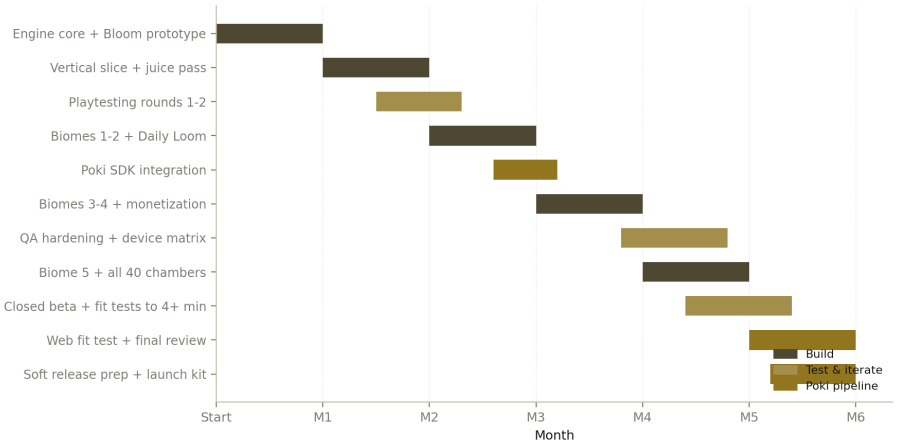


Figure 22.1 - The six-month plan: build, test, iterate, and the platform pipeline in the final month.

22.1 Phase plan and exit criteria

Phase	Window	Deliverables	Exit criteria (all must pass)
M1 - Foundation	Weeks 1-4	Core loop, hop movement, held singing, bloom wave shader, ink renderer, audio engine skeleton, verb system v1	A single test chamber is playable and beautiful; 60 fps on T2; first note sounds premium
M2 - Vertical slice	Weeks 5-8	Chambers C1-C5 with full juice; save system; adaptive score v1; playtesting integration	Two Poki playtest rounds complete; slice builds under 2 MB; the five-second wow confirmed in recordings
M3 - Content systems	Weeks 9-12	Biomes 1-2 (16 chambers), Daily Loom generator + validation, Songbook, SDK integration, events schema live	Daily generator proves 90 days solvable; SDK events verified in staging; C1-C16 smoke-tested
M4 - Depth and economy	Weeks 13-16	Biomes 3-4, rewarded placements, garden v1, cosmetics, quality tiers, device matrix pass	Full device matrix green; perf budgets hold on T3; economy plays coherently in internal runs
M5 - Complete and close	Weeks 17-21	Biome 5, all 40 chambers, share compositor, accessibility pass, closed beta (friends + itch build), first player fit tests	Player fit average at 4+ minutes (two consecutive tests); RC checklist green
M6 - Submission	Weeks 22-26	Final polish, thumbnail kit, submission with roadmap, playtest-to-fit-test iteration, web fit test, final review request	Web fit C2P 65%+; review requested; soft release prep complete

Table 22.1 - Phases, deliverables, and exit criteria

22.2 Risk-adjusted slack

The plan carries slack deliberately: month five contains two weeks of unbudgeted time whose only job is to absorb the two certainties of solo development - a hard technical surprise (the schedule's usual suspects: iOS audio behavior, and wave shader fill rate on T3) and a personal energy trough. If the slack survives unused, it converts to early submission; it is never converted to scope. The scope guard is absolute: the launch game is exactly the content in Chapter 7, and every good idea that arrives mid-production is written into the live-ops calendar instead of the launch build. The graveyard of web games is full of six-month plans that became fourteen-month plans one good idea at a time.

22.3 The submission month, week by week

Month six is not a development month; it is a measurement month, and its weeks are planned around the platform's own test cadence rather than around feature work. The rule for the month: no new content, only tuning, fixing, and testing - a submission built the same week it is uploaded is a submission built on hope.

- Week 22 - freeze and certify: RC build passes the full checklist; Poki Inspector clean; upload the candidate with thumbnail A; watch the first 10 playtest recordings the same day.
- Week 23 - first player fit tests: two tests on consecutive days, one build apart; read the histogram; fix only what the recordings prove; leave the build alone if the numbers hold.

- Week 24 - iterate to target: repeat the fit-test loop until average playtime sits at 4+ minutes with 30%+ over three minutes on two consecutive tests; begin the web fit test on Thursday to let it run over the weekend.
- Week 25 - web fit and final request: read CTR, time on page, C2P against category benchmarks; swap to thumbnail B if CTR scores under 3; on green numbers, request final review with the roadmap attached.
- Week 26 - buffer and soft-release prep: answer reviewer questions within a day; prepare the agreement information, the soft-release build flag, and the launch-week kit; if any target is still red, this week absorbs the iteration instead - the submission date bends, the quality bar does not.

23. The Solo + AI Production Model

This chapter defines how one person with a Claude subscription produces work that reads as a small studio's. The model has three non-negotiables: Claude is the production multiplier, the developer's taste is the quality system, and nothing ships unreviewed by a human being. Everything else - the workflows, the prompt recipes, the context discipline - exists to serve those three rules at a sustainable weekly rhythm.

23.1 The division of labor

Domain hat	Claude leads	Human leads	Cadence
Code (TS, shaders)	Drafting modules, refactors, test scaffolds, migration scripts	Architecture decisions, merge review, taste on feel code	Daily
Chamber content	Generating schema-valid candidates from pattern library + constraints	Playing every chamber; accepting 1 in 4; tuning	Daily in M3-M5
Design	Critique passes, option matrices, teardown summaries of reference games	All calls on tone, difficulty, fun	Weekly
Art / VFX	Shader math, atlas packing scripts, palette variant sweeps	The look; every curve and palette sign-off	Weekly
Audio	Parameter exploration briefs, recipe documentation, regression note-taking	All synthesis voicing; the mix; the identity	Weekly
QA	Test authoring, triage of playtest recordings into ranked hypotheses, release notes	Device sign-off, muted playthrough, RC checklist	Per build
Data	Funnel summaries, anomaly flags, cohort write-ups	The decision; what changes in the next build	Weekly in M5+
Writing / marketing	Devlog drafts, kit copy variants, community responses	Voice and final edits; the personal story	Per drop

Table 23.1 - Domain hats: who leads what

23.2 The working loop

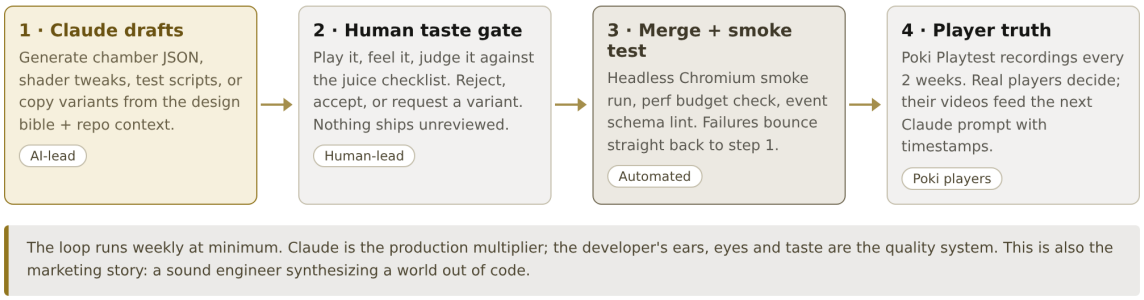


Figure 23.1 - The weekly production loop: Claude drafts, human taste gates, automation verifies, players adjudicate.

The loop runs on a simple rhythm: mornings are Claude-assisted generation blocks (code, chambers, tests) with the review gate applied immediately; afternoons are play-and-tune blocks where the developer's hands and ears do the work that cannot be delegated; Fridays are the review build, the smoke suites, and the week's funnel read. Every artifact that Claude produces lands in the repository through the same path as human work: a branch, a review, and the automated gates. The discipline is the moat - the flood of AI-slop games fails not because AI is weak but because nobody gates the output.

23.3 Context discipline

A persistent project context file (the design bible distilled from this PRD, kept under 500 lines) plus the repo's own structure gives every Claude session the same ground truth: the pillars, the verb table, the palette values, the event schema, the juice contract, and the current milestone. Sessions open with the bible and close with a diff summary appended to the worklog. This is unglamorous and it is the difference between an AI pipeline and an AI slot machine: the model's usefulness compounds with the quality of the context it is fed, and context quality is a habit, not a talent.

23.4 Prompt recipe library

Recipe 1 - Chamber generation

Chamber generation prompt

```
Generate ONE song chamber as JSON matching chamber.schema.json.
Constraints:
- Biome {biome}; pattern archetype: {pattern}
- Only verbs introduced by biome {biome} or earlier
- Solvable in {notes_target} +/- 1 notes; solve time 60-120 s
- One aha only; no fail state; no text labels
- Star thresholds: notes, time, paint - derive from your solve trace
Output: the JSON, then your own solver trace proving solvability,
then one line on what makes it interesting.
```

Recipe 2 - Juice review

Juice review prompt

```
Review this interaction spec against the juice contract (Table 9.1):
- Every input has response within 100 ms?
- Anticipation, squash/stretch, follow-through present?
- Audio + VFX responses specified, in key, at budget levels?
- Celebration escalates correctly vs neighboring events?
List violations as fixes, most severe first. No praise.
```

Recipe 3 - Playtest triage

Triage prompt

```
Input: timestamped notes from 10 playtest recordings.
Output a ranked hypothesis table:
1. Where players hesitated (timestamps -> chamber -> likely confusion)
2. Where they drifted (-> missing reward or challenge)
3. Top 3 fixes with expected funnel metric each will move
4. Which fix is cheapest to try first
Be specific: chamber IDs, seconds, and the exact frame you infer hesitation from.
```

Recipes 4-5 - Shader iteration and marketing copy

Compact recipes

```
Recipe 4 (shader): "Here is the wave fragment shader and a 3-frame behavior
description.
Goal: {effect}. Constraints: no branching per pixel, 16 instruction budget.
Propose the smallest change; explain the visual math in one paragraph."

Recipe 5 (copy): "Write 5 variants of {asset}: devlog title, store blurb,
creator-kit one-liner, daily-share caption, bug-fix tweet.
Voice: warm, specific, zero hype words. Keep each under {N} words."
```

23.5 The weekly operating rhythm

- Monday: read the week's funnel + recordings summary; set the week's single most important question.
- Tuesday-Thursday: two generation blocks and two tune blocks daily; every merge gated same-day.
- Friday: review build, smoke suites, RC-checklist delta, worklog and bible updates.
- Weekend: protected rest by policy - the plan's slack absorbs reality; burnout is a launch risk with a probability, not a moral failing.

24. Game Development Domain Handbook

A full studio is a stack of crafts: design, engineering, art, animation, audio, QA, data, production, marketing, community, and localization. This chapter is the handbook for every one of those crafts as they exist inside Songloom - what the discipline owns, its standard of quality here, its deliverables, and how solo-plus-AI covers it. It exists so that no domain is silently skipped just because there is no department to notice.

24.1 Game design

Owns the rules of play and the fun. Deliverables in this project: the verb system, the creature behaviors, the economy math, the difficulty doctrine, and the chamber grammar. Standard: every mechanic must serve a pillar (Chapter 5); every number must have a source (playtest or doctrine); every design decision must be reversible through data. Solo-plus-AI: Claude generates option matrices and critique passes; the human owns every call on tone and fun. Common failure mode avoided: feature accretion - mechanics that are clever but serve no pillar.

24.2 Level design

Owns the application of rules to spaces - the craft of teaching through arrangement. Deliverables: 40 chambers, the pattern library, star thresholds, shard placements. Standard: every chamber has goal, affordance, aha, and celebration; the solve is always visible in hindsight; difficulty rises by combination only. Solo-plus-AI: generation from the schema with a solver trace, human gate at 1-in-4 acceptance plus tuning. Failure mode avoided: chambers that test controller precision instead of thinking.

24.3 Narrative design

Owns meaning: the frame that makes mechanics matter. Songloom's narrative is minimal but real - a world that lost its song, creatures waiting to be woken, an Elderchoir that answers a player who sings well - and it is carried entirely by environment, melody, and creature behavior rather than text. Deliverables: the biome arcs, the Songbook melodies' titles, the Elderchoir call-and-response phrases, and the Daily Loom's day-number ritual. Standard: the story must be legible to a player who reads nothing; localization cost must remain zero by construction. Failure mode avoided: wordy lore that excludes the global audience.

24.4 Engineering

Owns correctness, performance, and maintainability. Deliverables: the modules in Table 12.1, the test suites, the build pipeline, the SDK adapter. Standard: strict TypeScript, zero runtime deps beyond renderer and SDK, budgets enforced by CI, determinism for replay and QA. Solo-plus-AI: Claude drafts and refactors; the human reviews every merge and owns architecture. Failure mode avoided: clever code that the game's future self cannot read - the bus factor is one, so readability is survival.

24.5 Art

Owns the visible world. Deliverables: the ink layer's geometry language, the bloom palette system, two texture atlases, creature silhouettes, the thumbnail kit, UI sheets. Standard: silhouette legibility at 64 px, hand-variance linework per the not-generated style guide, colorblind-safe verb palette. Solo-plus-AI: Claude writes shader math and packing scripts; the human owns every pixel of the look. Failure mode avoided: asset-store sameness - the single fastest tell of a generated game.

24.6 Animation and motion

Owns how the world moves. Deliverables: hop squash-and-stretch, creature idle and response sets, wave and reveal motion, UI transitions, celebration cascades. Standard: authored easing per object family (Chapter 10.5); anticipation before every action; nothing linear, nothing instant, nothing lazy. Solo-plus-AI: Claude proposes curve variants; the human tunes by eye in the tune blocks. Failure mode avoided: shared generic curves that make a hand-crafted game move like a template.

24.7 VFX and motion graphics

Owns spectacle: the bloom wave, particles, the paint persistence pass, and the share-card compositor - the surfaces where the game shows off. Deliverables: the wave shader, the particle library (motes, leaves, shards, shimmer), celebration choreography, share-card layouts. Standard: the wave is the signature; every VFX reads in a 2-second GIF; budgets cap at 220 alive particles. Solo-plus-AI: parameter sweeps generated, look gated by the human. Failure mode avoided: stock explosions and glow spam - the antithesis of the ink-and-bloom identity.

24.8 Audio engineering

Owns the game's soul. Deliverables: the five voices, 60+ event recipes, the adaptive score with four layers, the bus mix, latency handling, and the muted-playthrough parity. Standard: everything synthesized, everything in key, mix at minus 14 LUFS, the player's note always the soloist. Solo-plus-AI: documentation and regression support; the human voices every sound. Failure mode avoided: audio as decoration - here audio is the mechanic, and its engineering quality is the product.

24.9 Quality assurance

Owns the truth about the build. Deliverables: the device matrix, the smoke and perf suites, the schema and solver validation, the RC checklist, the muted and colorblind passes. Standard: no candidate reaches the platform without green CI and a manual matrix row. Solo-plus-AI: Claude authors tests and triages recordings; automation replaces the department. Failure mode avoided: testing what is easy instead of what breaks - devices, audio unlock, and save migration are where web games actually die.

24.10 Data and analytics

Owns learning from players. Deliverables: the event taxonomy, the metric targets and decision rules (Table 15.1), the weekly funnel read. Standard: every dashboard row maps to a pre-committed action; no vanity metrics. Solo-plus-AI: Claude writes the summaries and anomaly flags; the human decides. Failure mode avoided: data collection without decision - the analytics tax with no return.

24.11 Production and project management

Owns shipping. Deliverables: the roadmap in Chapter 22, the weekly rhythm, the scope guard, the risk register (Chapter 25), the worklog. Standard: every phase has exit criteria; slack is scheduled; scope is defended. Solo-plus-AI: Claude maintains status summaries from the worklog; the human holds the priorities. Failure mode avoided: the infinite project - scope discipline is the single strongest predictor of a solo game shipping at all.

24.12 Marketing, community, and localization

Owns the world's invitation. Deliverables: the devlog series, the creator kit, the community spaces and rituals, the reputation layer, and - by design rather than effort - a game that localizes itself through text-light visual design and a synthesized audio identity that carries across every market the platform serves. Standard: zero cash spend, presence over performance, reputation compounding. Solo-plus-AI: drafts and variants generated; the personal voice is never delegated, because the developer's story is the marketing. Failure mode avoided: launch-day silence - the kit and series are built in month five, before submission, not after.

25. Risk Management

The register below scores the project's real risks and pre-commits responses. Two structural facts shape it: the bus factor is one, and the single largest commercial risk named in the brief - rejection for insufficient uniqueness - is addressed not by hope but by the evidence dossier in Chapter 18. Scoring: likelihood and impact on a 1-5 scale, with 5x5 exposure.

Risk	L	I	Early signal	Mitigation (pre-committed)	Contingency
Poki rejection on originality	2	5	Reviewer questions during review stage	Evidence dossier (18.1); genre teardown (4.2); category strategy (18.2)	CrazyGames submission with same build; iterate and resubmit changed build
Player fit test below 4 min	3	4	First fit tests in M5	Session-0 doctrine; playtest cadence; chamber re-teach loop (15.3)	Delay submission one month; use scheduled slack; never submit weak
Wave shader fill rate on T3	3	3	Perf suite regressions in M2	Simplified wave path in Standard tier; early depth reject	Compatibility ink mode is fully playable; quality tier auto-drops
iOS audio unlock / latency edge cases	3	3	Device matrix rows in M2	First-tap unlock contract; measured latency constant (11.7)	Audio-visual parity ensures muted play never blocks progress
Daily generator produces dull chambers	3	3	Solver trace pass rate; beta feedback	Constrained grammar + 90-day validated buffer; human spot-review	Fall back to hand-authored dailies from a 60-chamber pool
Solo burnout / energy trough	3	4	Slippage two weeks running	Protected weekends; slack month five; scope guard absolute	Cut to 3 biomes at launch (30 chambers); live-ops completes the arc
Scope creep from good ideas	4	3	Backlog growin g inside a milestone	Live-ops capture rule: new ideas go to year-one calendar, never the launch build	n/a - the rule is the contingency
Cozy segment underperforms on web	2	4	Web fit test CTR in Creativ e/Puzzle below 3-score	Thumbnail kit A/B; category re-suggestion from four alternates	Lean into Skill category benchmarks; daily loop carries acquisition
Revenue below survival floor	3	3	First soft-releas e month under $500	Portrait + Gamebar passive revenue; update cadence earns promotion	Portfolio thesis: ship game two on the same engine; licensing conversations open at proven retention

Table 25.1 - Risk register

Two risks deserve standing attention beyond the table. Rejection risk is managed front-loaded: the concept was selected by whitespace analysis against the platform's own originality criterion, and the submission package carries the differentiation evidence. Burnout risk is managed honestly: the schedule carries slack, the scope guard is absolute, and the plan's fallback - shipping three biomes instead of five, with live-ops completing the arc - is a real product, not a failure mode. The graveyard of solo web games is not full of games that were too small; it is full of games that were never finished.

26. Success Metrics and Financial Outlook

Success is defined at three altitudes - the submission gates, the launch bars, and the 90-day trajectory - because a plan that only defines victory at the end cannot steer toward it. The financial outlook is calibrated to the platform's own published expectations for a first web game: five hundred to three thousand US dollars per month. The scenarios below are honest about that band and about what moves a game above it.

26.1 The KPI ladder

Stage	Metric set	Green
Submission gates (M5-M6)	Player fit avg; plays >3 min; C2P; benchmark scores	4+ min; 30%+; 65%+; 3+ scores in chosen categories
Launch bars (first 2 weeks)	C2P, playtime, first-session chambers, crash rate	70% C2P; 5+ min avg; 4+ chambers; under 0.5% sessions with errors
90-day trajectory	Daily Loom return, update lift, rewarded interact, promotion events	18%+ daily return; visible playtime lift per drop; 8-15% interact; platform promotion earned

Table 26.1 - Success definition at each stage

26.2 Revenue scenarios

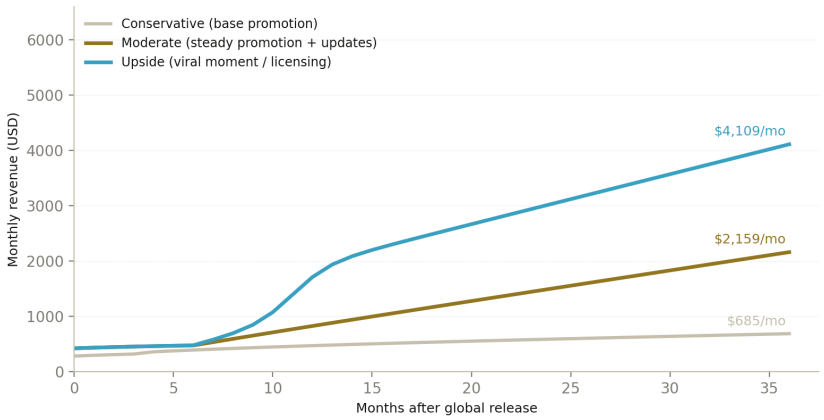


Figure 26.1 - Three revenue scenarios over 36 months: base, steady promotion with updates, and a viral or licensing upside event.

The conservative scenario assumes release with average promotion and a stable but modest audience: revenue in the several-hundreds monthly range, consistent with the platform's floor guidance for first games.

The moderate scenario is the plan's actual target: the update cadence earns continued promotion, the daily loop compounds retention, and revenue settles in the low thousands monthly. The upside scenario is not a plan but an option: a genuine viral moment around the daily loop or a creator clip, or a licensing conversation once retention is proven - both of which the design explicitly sets up (the share artifact and the platform's licensing posture) but neither of which the plan depends on.

26.3 Cost structure and the compounding thesis

~$0	~$20	1 dev
cash cost: tooling is free; Claude subscription only	monthly subscription already in the budget	plus an AI pipeline that drafts, tests, and documents

The financial structure is deliberately asymmetric: the cash at risk is effectively zero, the time at risk is six months of focused work, and the assets built - the engine, the chamber pipeline, the synth library, the distribution relationship - all survive into a second game at near-zero marginal cost. That is the compounding thesis for a solo web developer: the first game buys the pipeline, the second game arrives in half the time with an existing audience and a pre-trusted submission record, and the platform's own developer narratives - solo developers whose first hit rolled into a durable career - describe exactly this shape. Expectations are kept honest: the realistic near-term outcome is a modest but real income stream, a stronger portfolio, and a machine that makes the next game cheaper; the viral outcome is a bonus the design has earned the right to hope for, not one it needs to survive.

Appendix A: Poki Criteria Traceability Matrix

Every published platform criterion, mapped to the specific design decisions and document sections that satisfy it. This matrix is included in the submission package as the reviewer's shortest path to the evidence.

Poki criterion (published)	Design decision satisfying it	PRD section
Quality: UX feel, core loop potential	Single-verb bloom loop; 5-second wow; full juice contract with response matrix	6, 9.1-9.2
Quality: complete, not a prototype	40 chambers, daily system, garden, collection, cosmetics, live-ops calendar at submission	7, 20
Player fit: audience match	Cozy creativity segment (platform audience data); no-fail doctrine; text-light global design	3.1, 6.6, 24.3
Tech: load times	Progressive 3-stage loader; 1.5 MB first playable; real-byte progress UI	13.1
Tech: stable frame rates across devices	Fixed-step sim; quality tiers; device matrix sign-off; 30 fps T3 floor	12.3, 13.3, 16.1
Tech: lean file size	Zero audio files; two 512 px atlases; subset font; 6 MB total target vs 8 MB ceiling	13.1-13.4
Originality: fresh mechanic	Pitch-as-verb painting system with permanent world-state; no precedent found in web, cozy, or verb-based form	4.2-4.3, 5.1, 6.2-6.4
Depth and replayability	Loop opens over time: stars, shards, dailies, garden, mastery economy	7.4, 8.3
First impression: thumbnail	Three-thumbnail kit designed for 120 px category-page legibility	10.4
First impression: immediately playable opening	Menu-less boot into chamber 1; first input within 5 s; no-fail opening chambers	9.2, 8.3
Cross-device feel: thumb and mouse	One-thumb portrait-first control spec; keyboard parity; generous targets	6.1-6.2, 16.3
Room to grow: update systems	Data-driven chambers; monthly live-ops cadence; year-one roadmap in submission	12.2, 20, 22

Table A.1 - Criteria-to-design traceability

Appendix B: Chamber Content Specification (Samples)

Two complete sample chambers in the data format, demonstrating the pipeline end to end: the tutorial chamber that opens the game, and a mid-game duet chamber from the final biome. Both are schema-valid and solver-proven as specified in Chapter 12.

B.1 Whisper Hollow C1 - The First Song (tutorial)

wh_hollow_c01.json

```
{
  "id": "wh_hollow_c01",
  "biome": 1, "name": "The First Song",
  "size": [10, 14],
  "noFail": true,
  "anchors": [
    {"id":"spawn", "x":2, "y":12, "type":"ground"},
    {"id":"bell_ledge", "x":7, "y":4, "type":"ground"},
    {"id":"bloom_pad", "x":5, "y":9, "type":"ground"}
  ],
  "verbZones": [
    {"zone":"wall_main", "verb":"reveal", "paint": true},
    {"zone":"door_seal", "verb":"mid_charm", "target":"bloomling_1"}
  ],
  "creatures": [
    {"type":"bloomling", "id":"bloomling_1", "at":"bell_ledge", "carries":"mid"}
  ],
  "goal": {"type":"bloom_door", "requires":["bloomling_1.charmed"]},
  "teach": {"verb":"mid", "hint":"lantern_pulse_near_bloomling"},
  "stars": {"notes":2, "time_s":35, "paint_pct":70},
  "events": ["level","wh_hollow_c01","start|complete|fail"],
  "audio": {"motif":"hollow_owl", "key":"D_dorian"}
}
```

Play script: the Hummy wakes mid-room; the world is ink and paper. The first tap anywhere produces the first chirp - the five-second wow - revealing the wall's fern ticks. The lantern heart pulses amber when the player's held note sweeps near mid pitch, teaching the voice without a word. The Bloomling on the bell ledge stirs at the wave's edge, wakes, and hums the door open; the celebration paints the room amber from the door outward. Two notes, no possible failure, thirty-five generous seconds: the product's entire thesis in one chamber.

B.2 Elderchoir Keep C7 - The Answering Stone (duet)

ec_keep_c07.json

```
{
  "id": "ec_keep_c07",
  "biome": 5, "name": "The Answering Stone",
  "size": [16, 22],
  "noFail": false,
  "anchors": [
    {"id":"spawn","x":2,"y":20,"type":"ground"},
    {"id":"duet_a","x":6,"y":15,"type":"duet_stone","voice":"low"},
    {"id":"duet_b","x":11,"y":9,"type":"duet_stone","voice":"high"},
    {"id":"moss_pocket","x":9,"y":12,"type":"echo_moss"},
    {"id":"exit","x":14,"y":3,"type":"ground"}
  ],
  "verbZones": [
    {"zone":"gap_a","verb":"grow","target":"bridge_a","maturity":2.0},
    {"zone":"crystal_shards","verb":"shatter","removes":"block_c"}
  ],
  "creatures": [
    {"type":"hushling","id":"hush_1","at":"moss_pocket","radius":3}
  ],
  "goal": {"type":"elder_door", "requires":["duet_sequence(low,high)"]},
  "stars": {"notes":4, "time_s":95, "paint_pct":80},
  "events": ["level","ec_keep_c07","start|complete|fail"],
  "audio": {"motif":"choir_call", "key":"G_major", "finaleLayer":true}
}
```

Play script: a chamber that audits the whole game. The player grows the bridge while the Hushling sleeps near the moss pocket, shatters the crystal block with a chime - waking the Hushling, whose drift must be routed with a deliberate silence - and then sings the duet: the low stone first, the high stone second, in order, or the Elder door stays sealed and the sequence resets softly. Four voices, one of every verb, and a finale that harmonizes with the player's own melody when the door opens. This is the shape mastery takes in Songloom: not harder inputs, but better listening.

Appendix C: Glossary and Sources

C.1 Glossary

Term	Meaning
Bloom wave	The colored ring emitted by a sung note; reveals geometry and paints it permanently
Verb (voice)	The world action bound to a pitch band: grow, charm, shatter, open, hide
Resonance	The chamber-level score expressing paint coverage and musicality; rewards, never gates
Hummy	The player creature; lantern heart previews pitch color and shows resonance
Bloomling	Helper creature woken by mid notes; carries notes to remote switches

Term	Meaning
Hushling	Soft obstacle that drifts toward sustained sound; sleeps in silence
Echo Moss	Relay surface that re-emits any wave from its own position
Elderchoir	Gatekeeper statues that open for chords and duet sequences
Harmony Notes	The single soft currency, earned in play, spent on cosmetics
Song chamber	A self-contained three-minute puzzle room; the core content unit
Daily Loom	The seeded daily chamber identical worldwide; the retention and share engine
Song Card	The spoiler-free share image of a player's painted daily result
Bloom Garden	The persistent no-goal creative space; cosmetics home
Player fit test	Platform test: 500 real players, playtime histogram; healthy = 3+ min average
Web fit test	Platform test on real category pages: CTR, time on page, C2P vs benchmarks
C2P	Conversion to play: share of visitors reaching the first gameplayStart event
Gamebar ads	Platform mobile ad format in the title bar; requires portrait support
Vertical layering	Adaptive music technique: layers added or removed against a fixed harmonic grid
T1/T2/T3	Device quality tiers; Standard (T3) is the performance floor for sign-off
Stage 1/2/3	Progressive loading bundles; stage 1 is first-playable under 1.5 MB
RC checklist	Release-candidate sign-off list in Chapter 16.4
Design bible	The 500-line living context distilled from this PRD that feeds every AI session

Table C.1 - Project glossary

C.2 Primary research sources

- Poki Documentation - What we look for; Final review; How testing works; Player fit test; Web fit test; Engagement; Easy access and onboarding; Monetization tips; Game Events; Post-release updates; Playtesting; Choosing your web game engine (developers.poki.com, accessed October 2026).
- Poki for Developers platform pages - scale, tooling, and partnership statements (developers.poki.com, 2026).
- Poki press and study coverage - 625 million players milestone (March 2026); 2026 State of Web Gaming study of 2,000 gamers and 400 developers (June 2026).
- IndieGameBusiness - Web Gaming for Indie Developers: 5 Honest Truths (April 2026), interview with Poki developer relations on review criteria, revenue expectations, and audience behavior.
- Poki Engineering - What it takes to build web games in 2026 (June 2026), including WebGPU coverage and file-size guidance.

- The Business Research Company - Browser Games Global Market Report (2026 edition): market size and 2030 projection.
- Industry coverage of the AI-slop problem in 2026 game submissions; cozy-gaming trend analyses; Wordle share-mechanic postmortems; Friday Night Funkin and .io phenomenon histories; Level Devil creator interviews.
- Platform catalog inspection - category saturation and current popular lists on poki.com (October 2026).